



Data Warehouses

Lecture 5



Reminder

Last meeting – SQL FOR OLAP:

Case, Pivot/UnPivot, Grouping Sets

Window

*OVER clause with PARTITION / ORDER /
ROWS/RANGE*

Execution plans in SQL Server

Lecture Goals

Goal:

BI – DataInformationKnowledge

High-level approaches

Data warehouses

Business Intelligence

Business Intelligences:

“refers to a set of tools and techniques that enable a company to transform its business data into timely and accurate information for the decisional process, to be made available to the right persons in the most suitable form.”

Encyclopedia of Database Systems

Idea

BI system

designed with the primary goal of extracting important data from an organization's raw data to reveal insights to help a business make faster and more accurate decisions

BI can succeed only as a result of cooperation between

technical developers and their business clients

Business Intelligence

BI environment is used by decision makers to:

Understand

the status of the organization

Collaborate

*on a shared view of data, business
and presentation logic*

Reduce

the time to decision

Goal is to:

Allow for measuring specific
operations

Develop an integrated perspective

Enhance competitive advantage

One of the more common ways to
present business intelligence is
through data visualization.

DATA



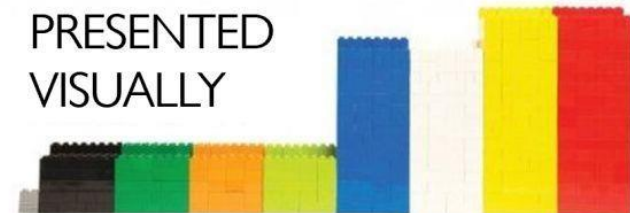
SORTED



ARRANGED



PRESENTED
VISUALLY



An abstract digital cityscape composed of glowing blue cubes and rectangular blocks. The surfaces of these blocks are covered in a dense pattern of binary code (0s and 1s). Several bright blue and green light sources are visible, some appearing as small dots and others as larger, glowing rectangular areas within the blocks. The overall color scheme is dominated by deep blues and greens, with occasional red highlights.

Business Intelligence

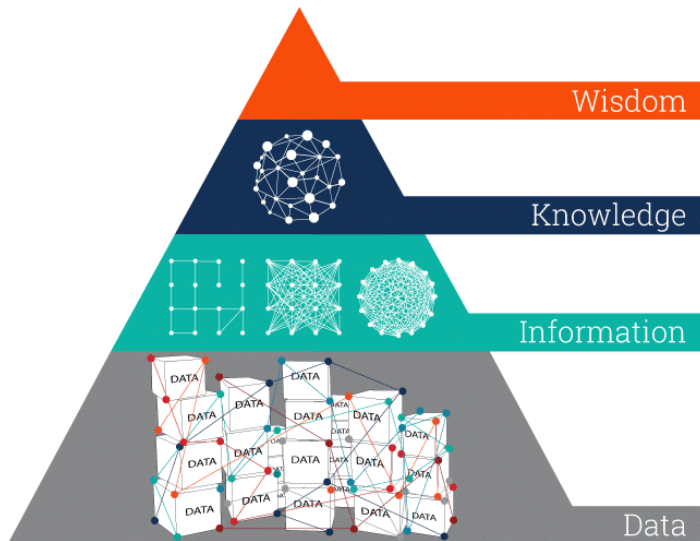
The Data Warehousing Institute defines BI as:

The processes, technologies,
and tools needed to

*turn data into information,
information into knowledge,
and knowledge into plans*

that drive profitable business
action.

Information asset



Each step up
the pyramid
answers
questions
about and
adds value
to the initial data.

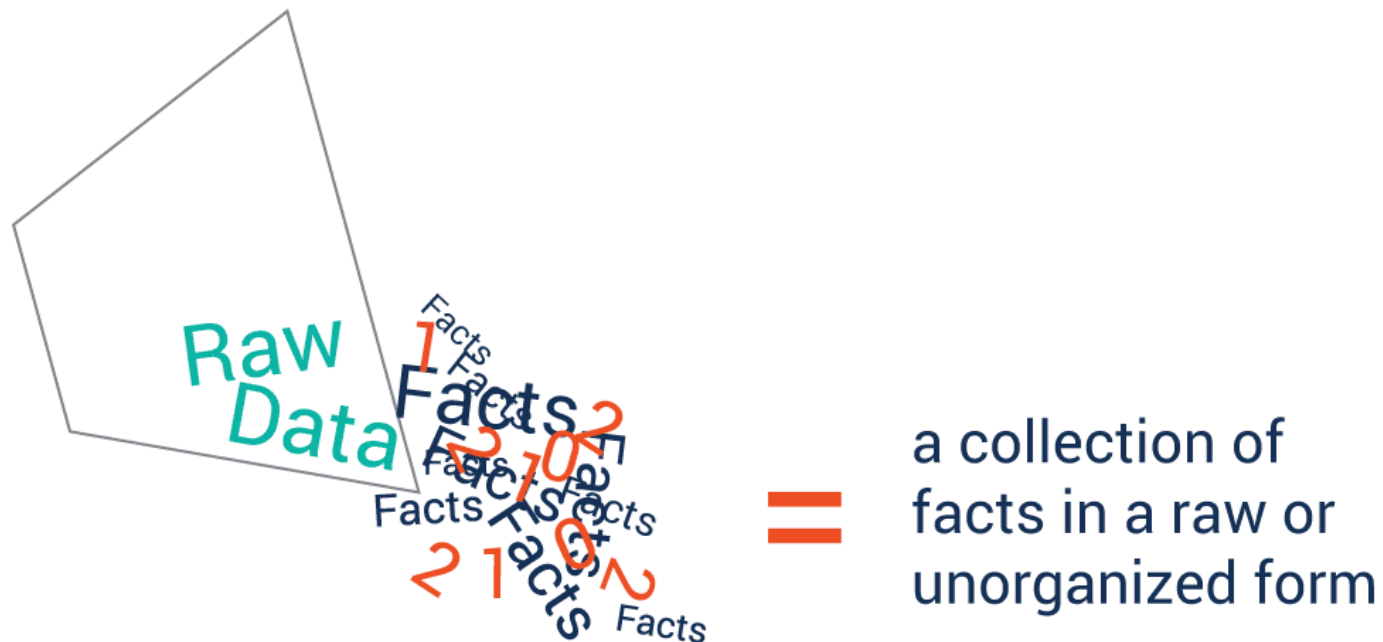
The data-information-knowledge-wisdom (DIKW) hierarchy

“Typically, information is defined in terms of data, knowledge in terms of information, and wisdom in terms of knowledge.”

Jennifer Rowley (2007)

We will focus on DIK part

Data / Information / Knowledge



Base building block - Raw **Data**

Data / Information / Knowledge

Data is raw, random, and unorganized.

Data is just a set of signals or symbols – that represent properties of objects, events and their environments.

It may be server logs, user behavior events, or any other data set.

The prime example of data and data acquisition is provided by automatic instrument systems; an unmanned weather station, for instance, may record daily maximum and minimum temperatures

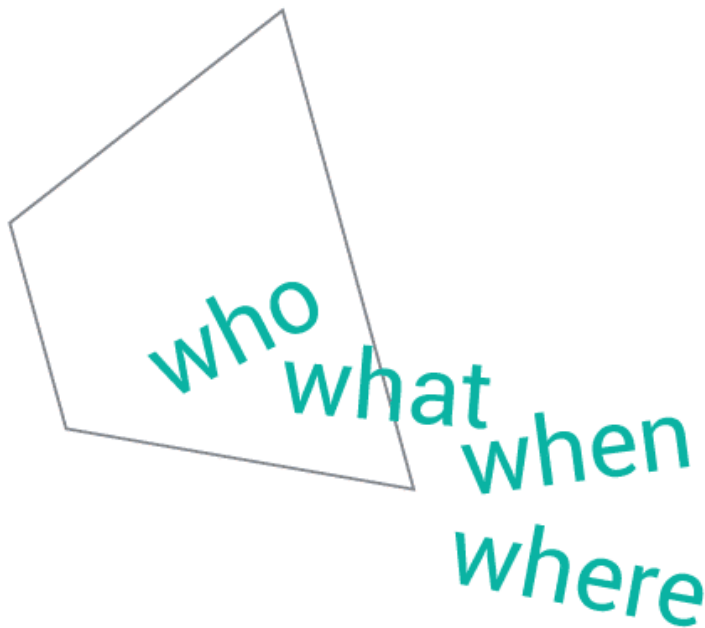
such recordings are data

It's unorganized and unprocessed (J. Rowley).

Information is data that has been organized, structured, and processed.

Information is what you use to gain knowledge.

Data / Information / Knowledge



easier to measure,
visualize and analyze
data for a specific purpose

Second building block - Derived **Information**

Data / Information / Knowledge

Data is raw, random, and unorganized.

Information is data that has been organized, structured, and processed.

You get Information when you start to make data useful

Data itself is of no value until it is transformed into a relevant form

This is relevant, or usable, or significant, or meaningful, or processed, data

In short, information is data with meaning.

For example, were an enquiry to be “what is the average temperature for July?”

there may be individual temperatures explicitly recorded as data, but perhaps not the average temperature; however, the average temperature can be calculated or inferred from the data about individual temperatures.

Information is what you use to gain knowledge.

Data / Information / Knowledge



Data / Information / Knowledge

Data is raw, random, and unorganized.

Information is data that has been organized, structured, and processed.

Information is what you use to gain knowledge.

Users of this hierarchy often construe knowledge as know-how, or skill, rather than knowledge in the sense of the know-that of propositional knowledge

Knowledge is know-how, for example, how a system works. It is what makes possible the transformation of information into instructions. It makes control of a system possible. To control a system is to make it work efficiently.

It implicitly requires learning.

It means that we can take data, categorize and process it generating Information, then organize all this Information in a way that it can be useful.

Data / Information / Knowledge

Data

a collection of raw value elements or facts used for calculating, reasoning, or measuring; may be collected, stored, or processed but not put into a context from which any meaning can be inferred.

"unstructured facts and figures that have the least impact on the typical manager"
(Thierauf, 1999)

Information

the result of collecting and organizing data in way that establishes relationships between data items, which thereby provides context and meaning.

for data to become information, it must be contextualized, categorized, calculated and condensed (Davenport & Prusak, 2000)

Knowledge

the concept of understanding information based on recognized patterns in a way that provides insight to information.

is closely linked to doing and implies know-how and understanding (Davenport & Prusak 2000)

Data / Information / Knowledge

Data

a collection of ingredients sitting on the counter, they include carrots, onions, leeks, garlic, and potatoes from the farmer's market, and a package of chicken, a box of rice, and some cans of broth from the grocery store.

In the data warehousing (DW)/BI world, this is like source data from different operational systems.

Information:

then you get everything ready by washing, peeling, and cutting up the vegetables, cutting up the chicken, and opening the cans of broth, you put it all in the pot and turn on the heat where it cooks and becomes soup.

In the DW/BI world, the data has been moved into the ETL (extract, transform, and load) system and is transformed into information.

Knowledge:

Now the soup is ready to be put into bowls and eaten

In the DW/BI world business people consume the information in reports to gain knowledge that helps them make informed business decisions

Data / Information / Knowledge

2.4.2 How are data, information and knowledge linked?

If we put Knowledge into an equation it would look like this:

Information + application or use = Knowledge

Example

Looking at the examples given for **data**:

- 3, 6, 9, 12
- cat, dog, gerbil, rabbit, cockatoo
- 161.2, 175.3, 166.4, 164.7, 169.3

Only when we assign a context or meaning does the data become **information**. It all becomes meaningful when we are told:

- 3, 6, 9 and 12 are the first four answers in the 3 x table
- cat, dog, gerbil, rabbit, cockatoo is a list of household pets
- 161.2, 175.3, 166.4, 164.7, 169.3 are the heights of the five tallest 15-year-old students in a class.

If we now apply this information to gain further **knowledge** we could say that:

- 4, 8, 12 and 16 are the first four answers in the 4 x table (because the 3 x table starts at three and goes up in threes the 4 x table must start at four and go up in fours)
- The tallest student is 175.3cm.
- A lion is not a household pet as it is not in the list and it lives in the wild.



Turning Knowledge into Actionable Plans

Actionable Knowledge

Being able to act based on the intelligence that we have learned is the key point of any BI strategy

If you are using BI for micromarketing,
finding the right customer for your product is irrelevant if you do not have a plan to contact that customer.

If you are using BI for fraud detection,
finding the pattern of fraud is of little value if your organization does not do something to prevent that fraudulent behavior.

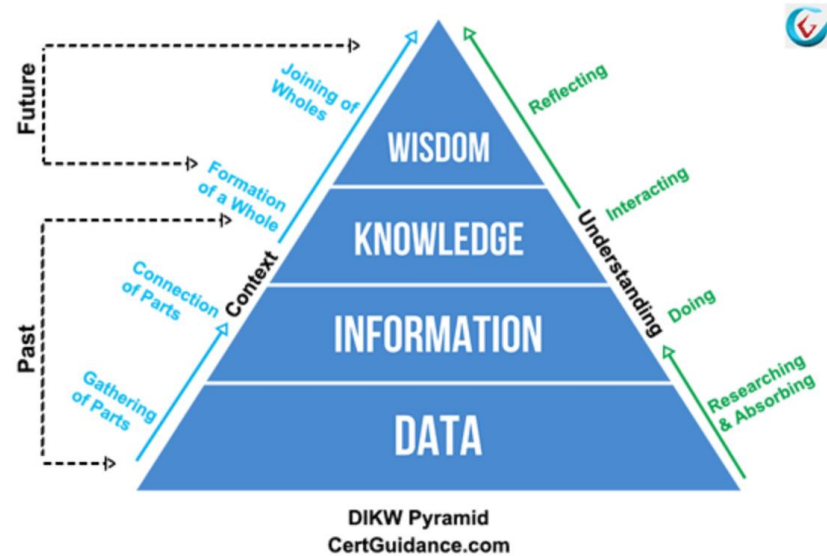
Data / Information / Knowledge

Two different angles:

As per the contextual concept,
moves from a phase of gathering data parts (data), the connection of raw data parts (information), formation of whole meaningful contents (knowledge) and conceptualize and joining those whole meaningful contents (wisdom).

From the understanding perspective

as a process starting with researching & absorbing, doing, interacting, and reflecting

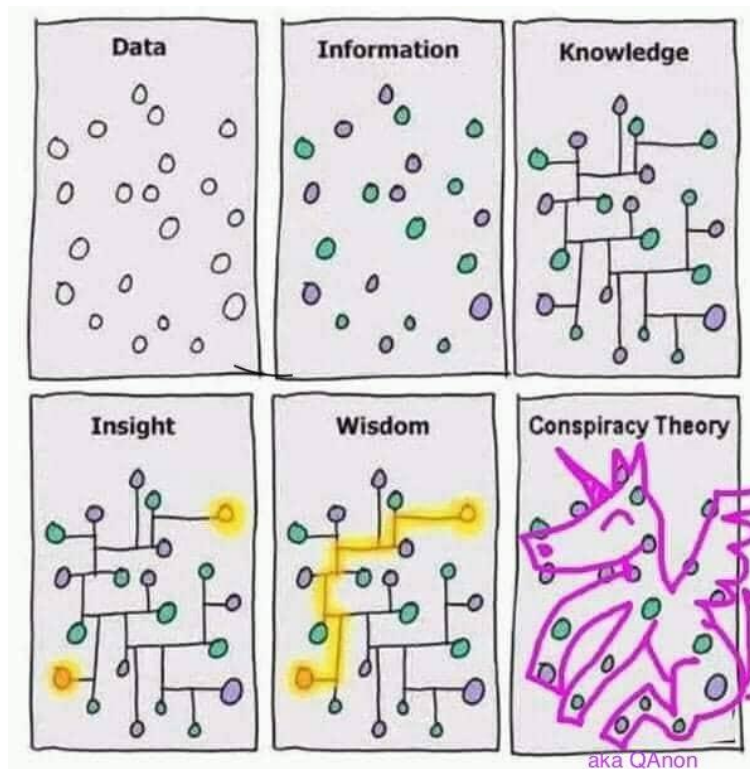


Data / Information / Knowledge



The top of the DIKW hierarchy - Guiding **Wisdom**

Data / Information / Knowledge





High level approaches

Approach

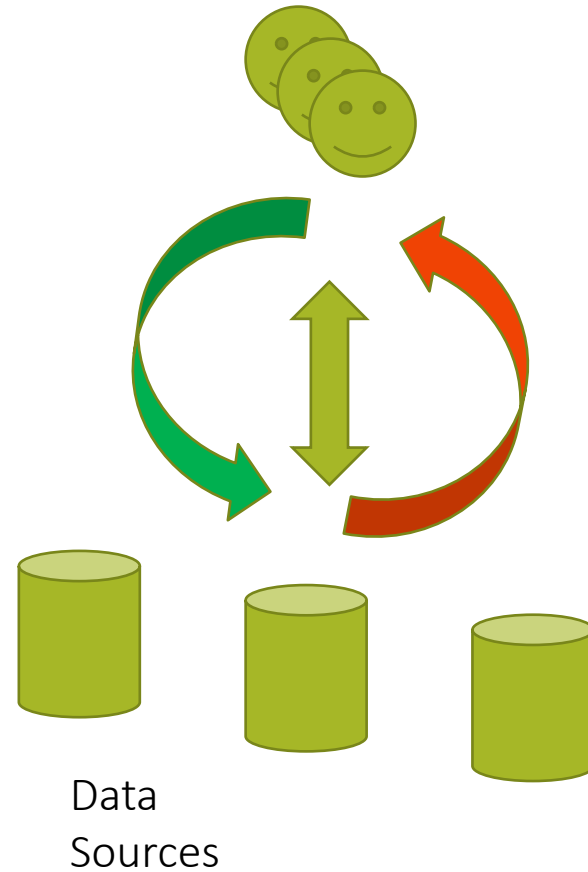
Situation

multiple heterogeneous
databases

operational

query requirements

analytic



Data

Available data:

operational data:

Legacy systems,

Dedicated applications

Enterprise applications – ERP, MRP II, CRM (OLTP)

databases:

relational, non-relational

data dumps

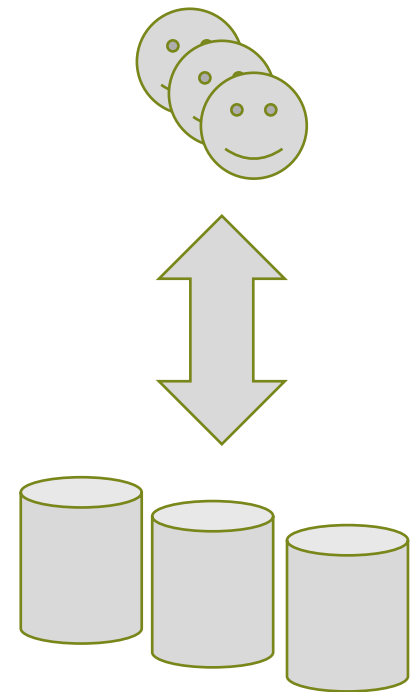
general data:

external sources, e.g., web services

archives:

external data storages,

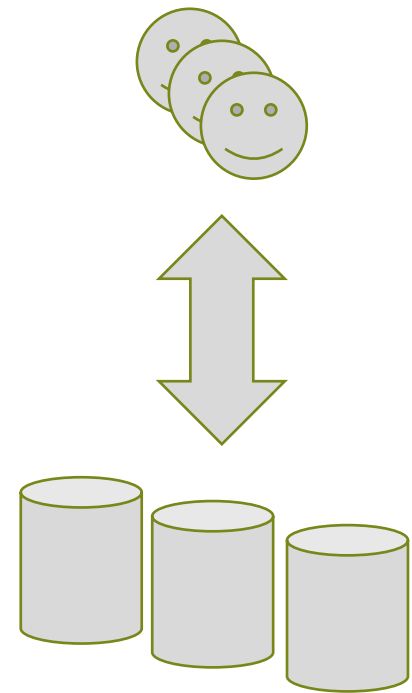
removable media



Data

Available data:

- different locations
- different formats and structures
- different regulations and policies
- different availability
- different semantics
- different quality standards



A scenario,

The management of a large retail chain operating from multiple locations

Each, such, branch generates massive amounts of data.

Assume: at the end of day, want to know about transactions done that day.

One could take the case of a Dominos Pizza store, where the management needs to know the total sales for that day or other details such as the number and types of pizzas sold.

Such companies rely on OLTP systems to get data from multiple stores spanning the world.

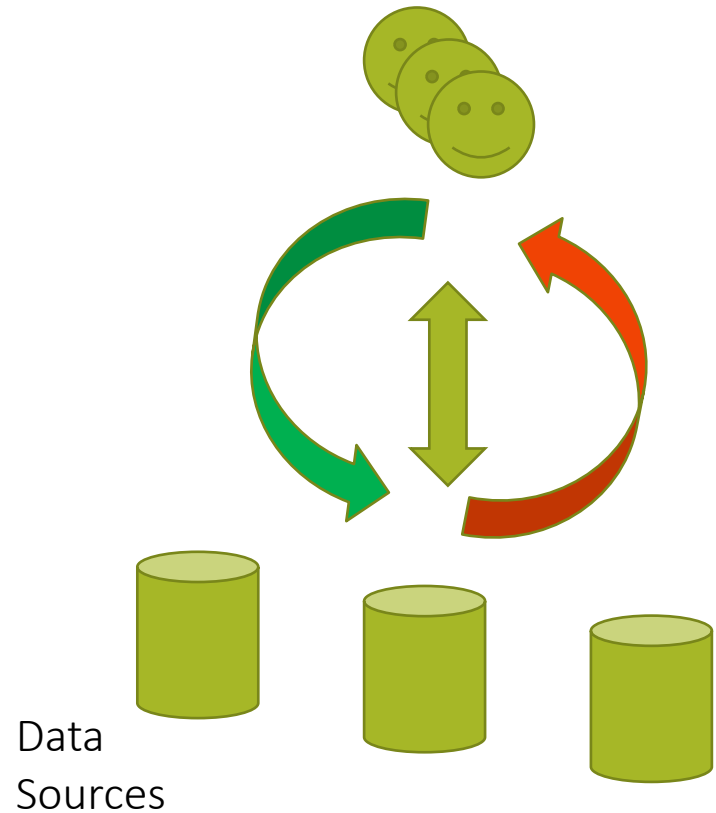
On these OLTP systems, queries usually run on an indexed database, as this makes the searching fast and efficient.

But unfortunately, data spread over multiple systems leads to a plethora of technical errors when carrying out the simple task of running queries on data stored on OLTP systems.

Approach

How to handle queries?

How to handle responses?



Query Driven Approach

Query Driven Approach

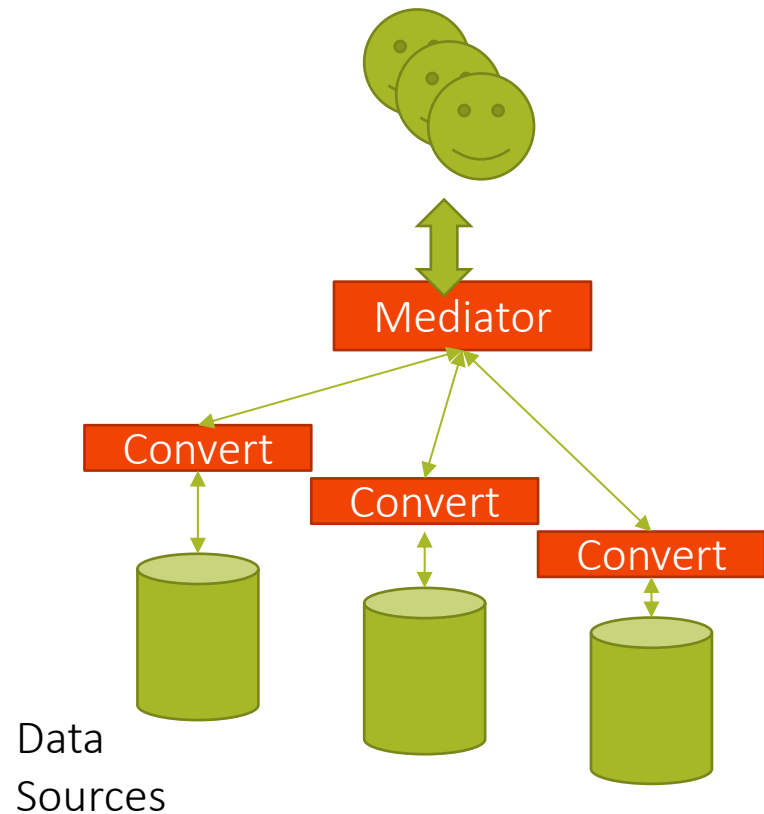
build wrappers and integrators

to integrate heterogeneous databases

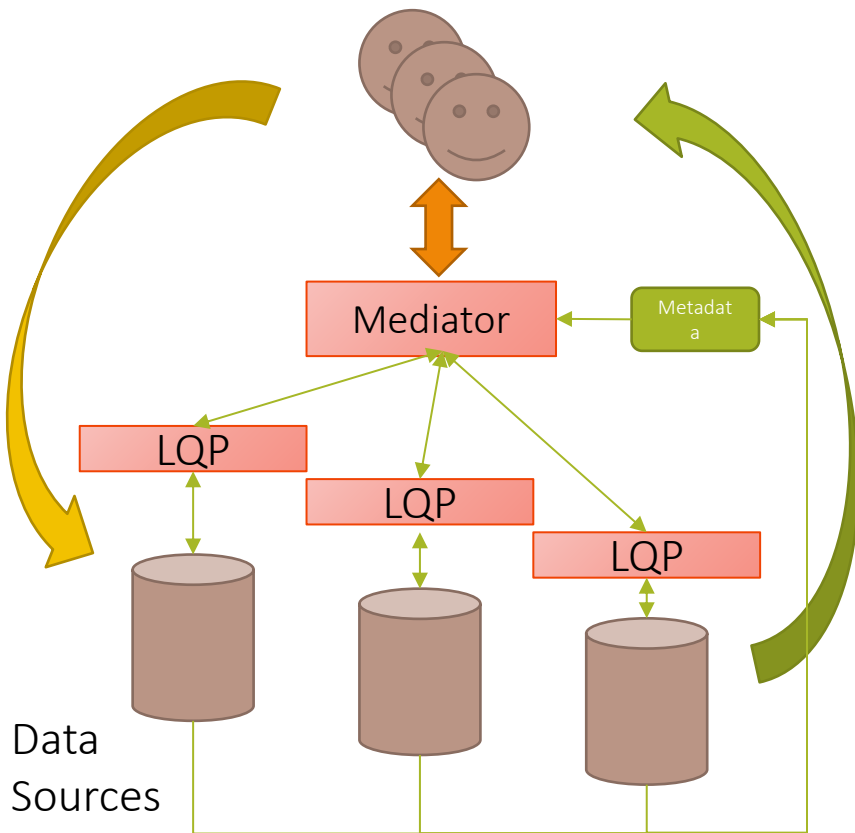
wrapper – a tool to access known resources and translate their objects, focused on making an interface compatible

mediator – unifies several interfaces, notably their way of communicating, like an actual mediator.

lazy integration



Query Driven Approach



Query Driven Approach

Lazy, on-demand

query is issued to a client side

*a metadata dictionary translates
the query into inner queries*

queries are mapped and sent to
the local query processor

*inner queries are appropriate for
the individual heterogeneous sites*

results from heterogeneous sites
are integrated into a global
answer set

Query Driven Approach

Query Driven Approach

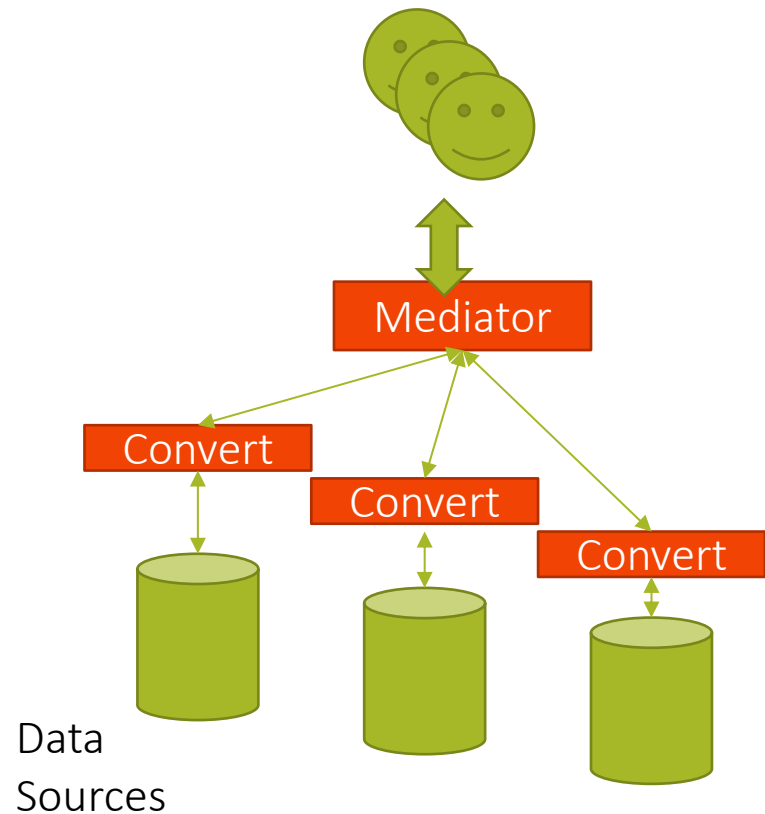
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lazy integration



Query Driven Approach

On demand query

management queries are run on OLTP systems in real-time

as soon as data is received, the queries are run giving the management the advantage of up-to-date information.

due to the complex nature of queries, on demand query approach loads OLTP systems heavily and hence, is unacceptable in real life scenarios.

Eager approach

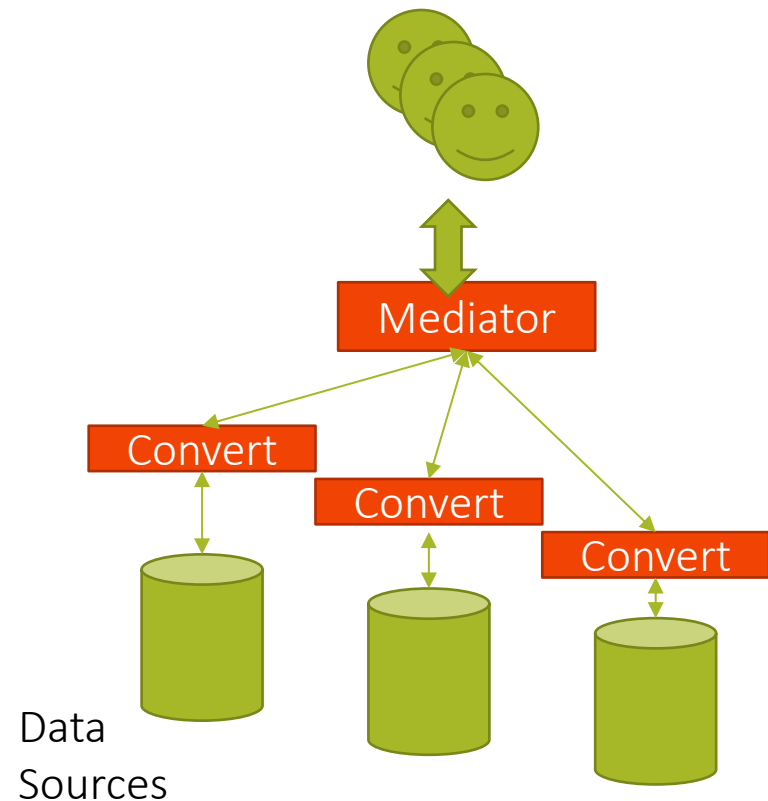
management queries are collected on a regular basis, and during nonpeak hours these queries are run on the OLTP systems.

results of these queries are stored separately, whenever a request from management is found, the saved query results are provided.

is quite quick

inability to deliver up to date information.

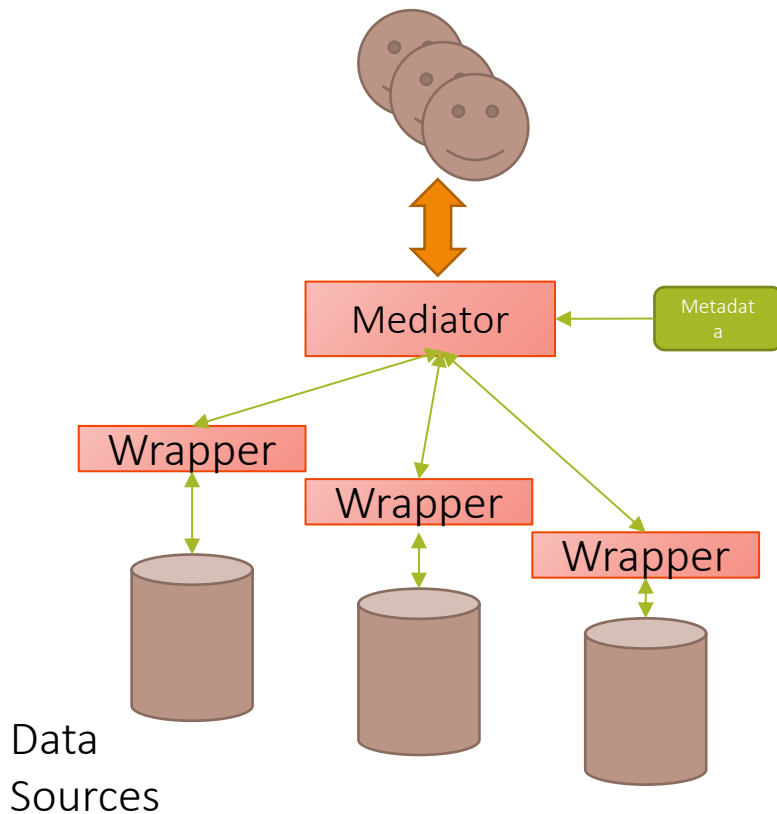
As queries are not run-on real-time data it carries data of some previous state.



Query Driven Approach

Query Driven Approach

Advantages / Disadvantages ?



Query Driven Approach

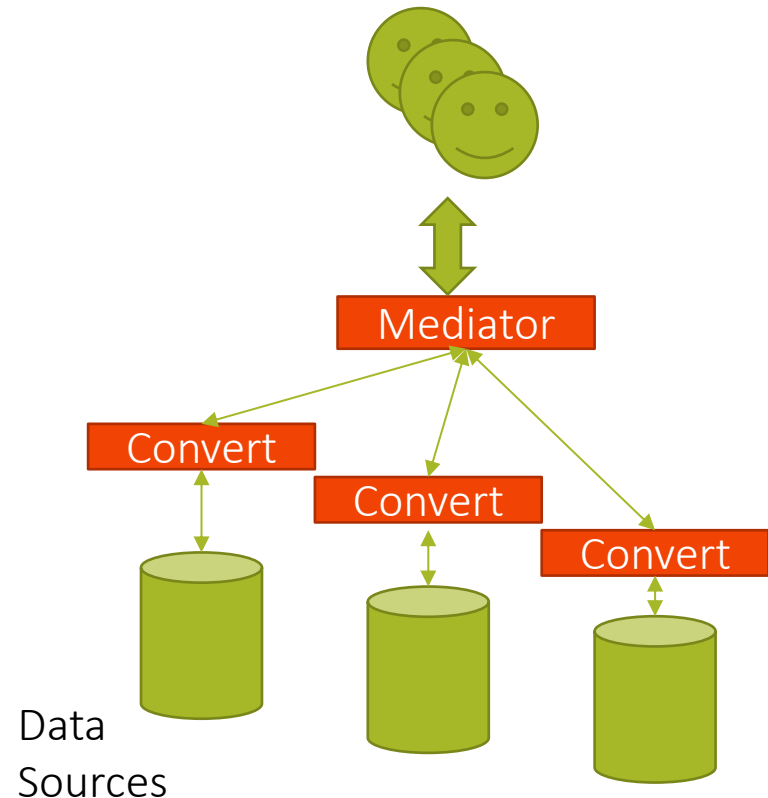
Query Driven Approach

Disadvantages

needs complex integration and filtering processes

is very inefficient and very expensive for frequent queries

very expensive for queries that requires aggregations



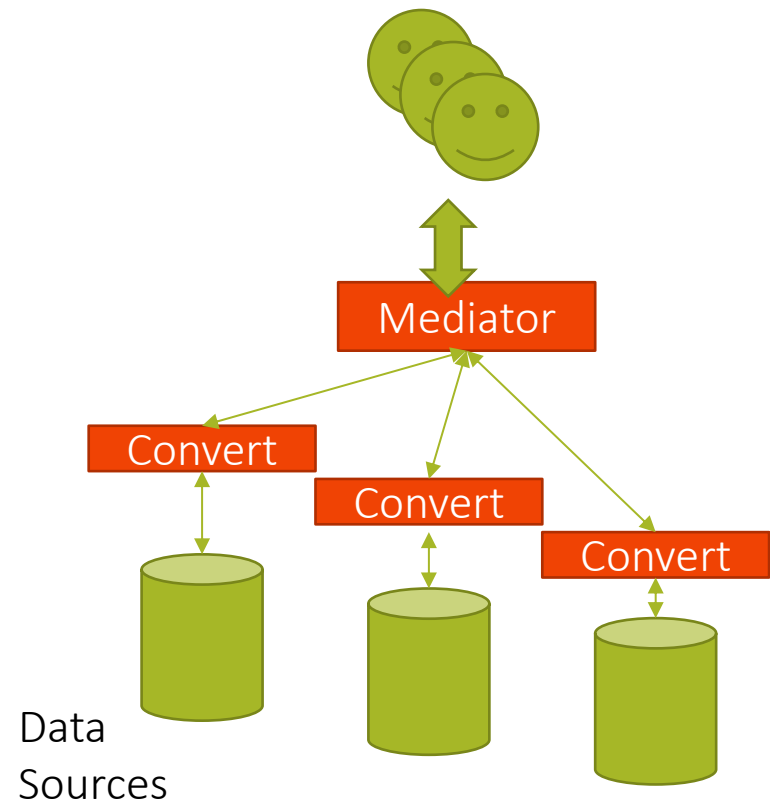
Query Driven Approach

Query Driven Approach

Advantages

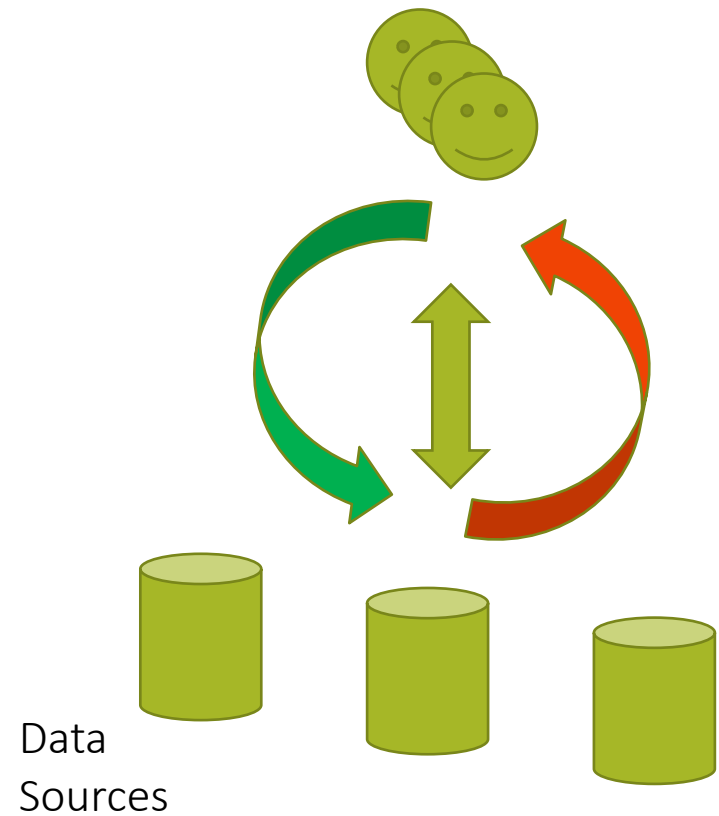
access to current data

no data redundancy

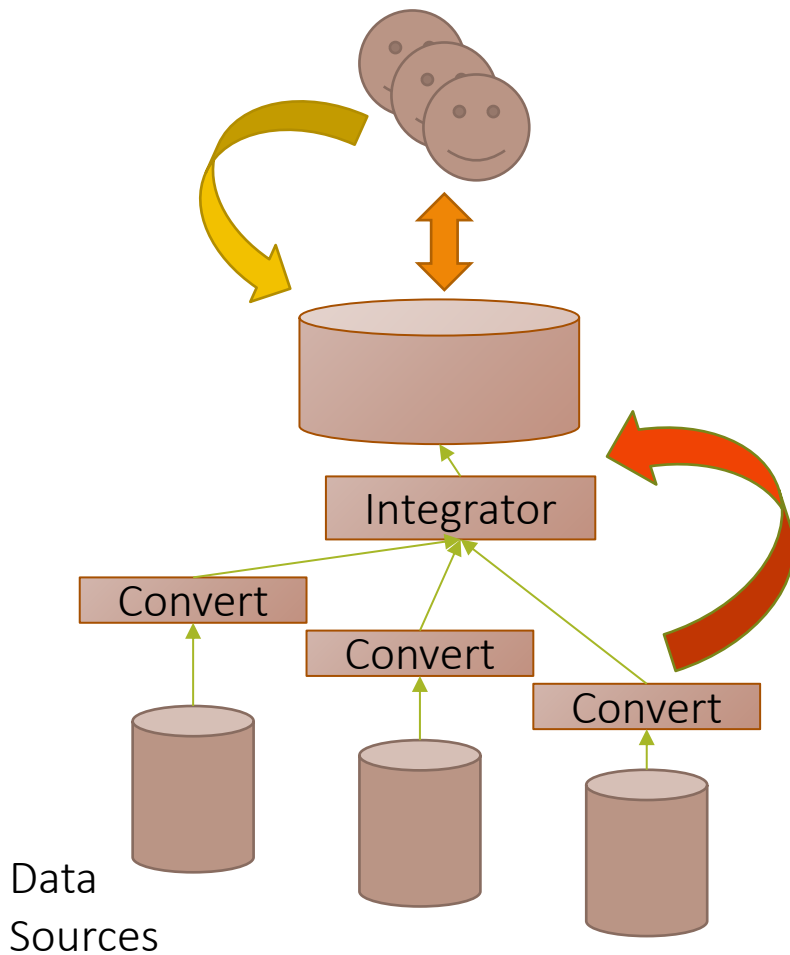


Approach

Different approach



DW Incentives

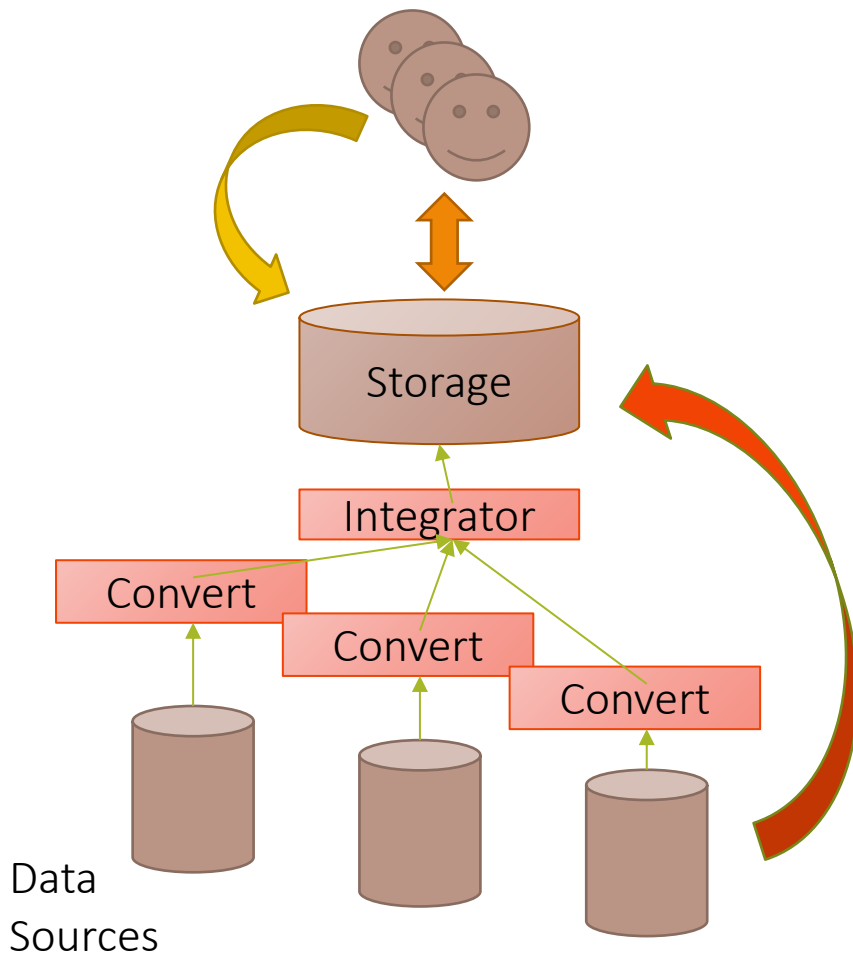


Update Driven Approach

the information is integrated
*from multiple heterogeneous
sources*

in advance and stored
information is made available
for direct querying and analysis

Update driven approach



Update Driven Approach

Proactive, not even eager
the information is integrated
*from multiple heterogeneous
sources*

in advance and stored
information is made available
for direct querying and analysis

Update Driven Approach

Why a separate storage?

High performance for both systems

Transactional layer

tuned for OLTP:

access methods, indexing, concurrency control, recovery

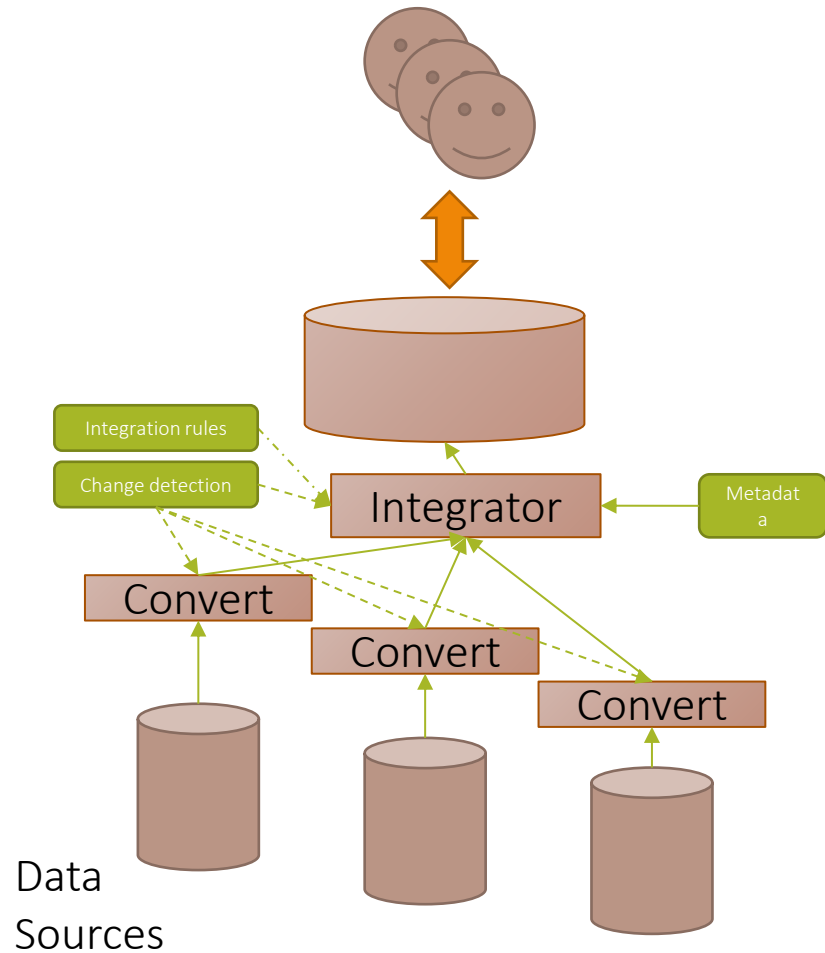
Analytical layer

tuned for OLAP:

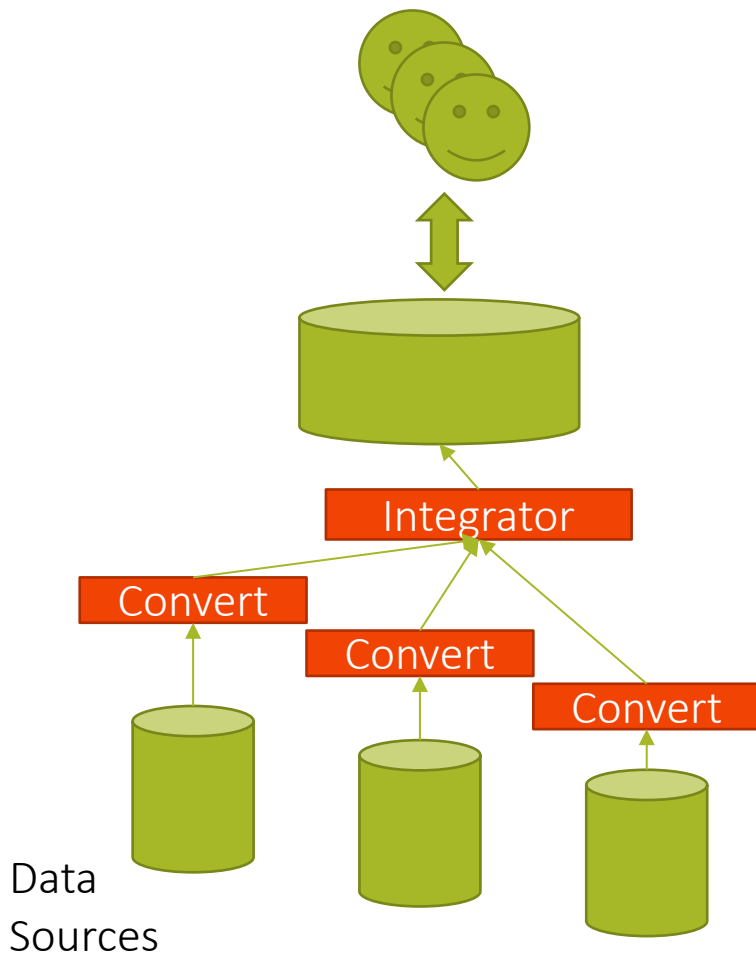
complex OLAP queries, multidimensional view, consolidation

DW Incentives

Update Driven Approach
Advantages / Disadvantages ?



Update Driven Approach



Update Driven Approach

Advantages

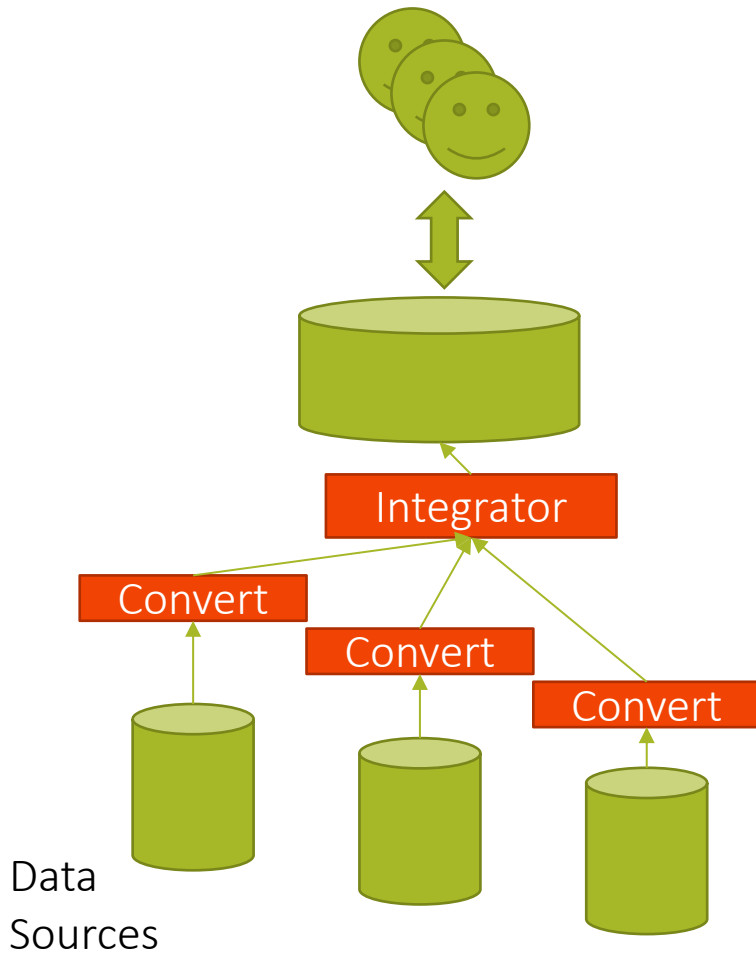
provide high performance

data are copied, processed, integrated, annotated, summarized and restructured

in semantic data store in advance

query processing does not require interface with the processing at local sources

Update Driven Approach



Update Driven Approach

Disadvantages

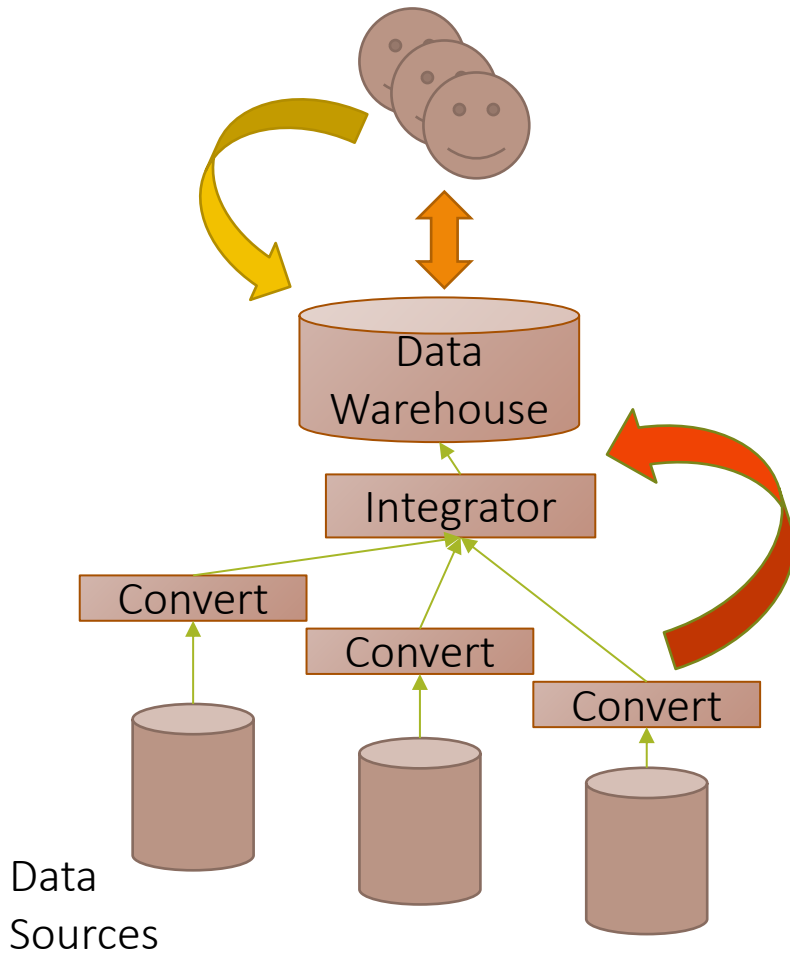
High data redundancy

Problems with update

Update Driven Approach

Update Driven Approach

Data Warehouse as an analytical data repository



BI Process

Data from the operational systems are

- Extracted

- Cleansed

- Transformed

- Aggregated

- Loaded into the DW

A good DW is a prerequisite for a successful BI solution



Data in DW

Data in DW:

replicated, cleansed, transformed

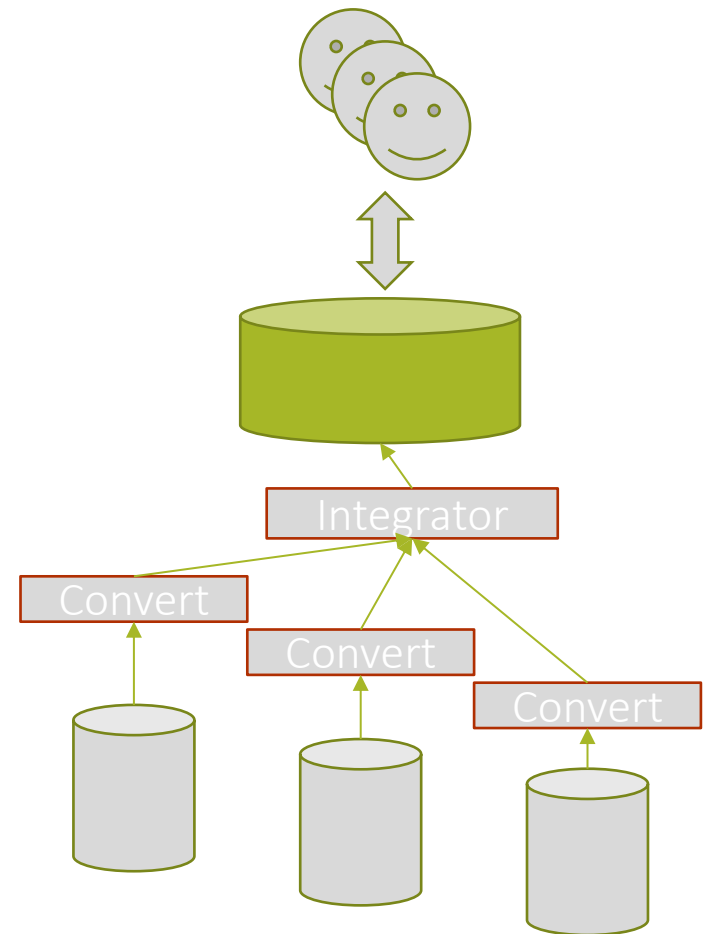
aggregated (pre-aggregated)

loaded to the data warehouse system

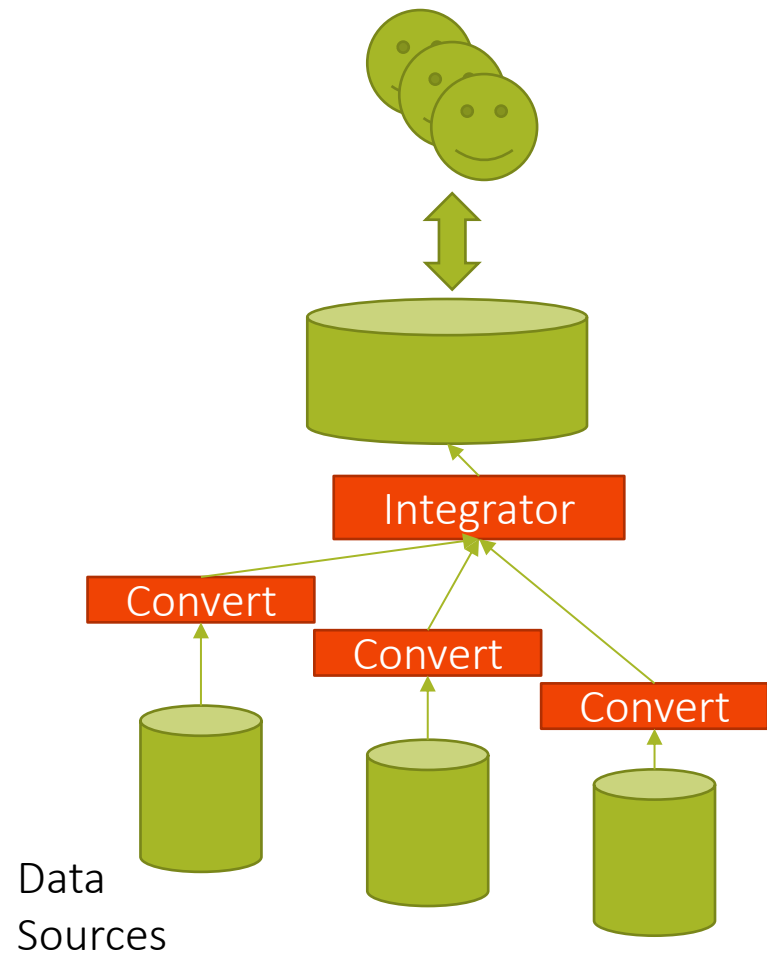
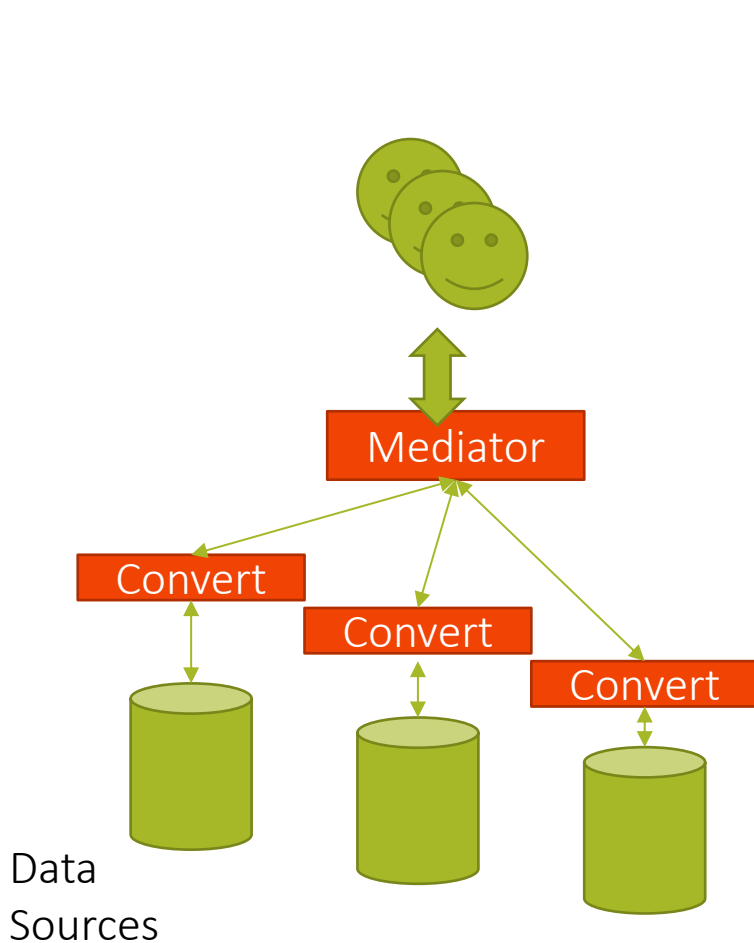
as opposed to CrUD

refreshed

*managed process; nearly never a data warehouse
is a one-time endeavour*



Query / Update Driven Approach





Data Warehouse

<https://www.youtube.com/watch?v=rvURMymCpJM>

DW Incentives

Incentives for a Data Warehouse

Businesses have a lot of data, operational data and facts.

data is usually in different databases and in different physical places

distributed

data is available (or archived), but in different formats

heterogeneous

Decision makers need

to access information

(data that has been summarized) virtually on one single site.

this access needs to be fast

regardless of the size of the data, and how old the data is.



Data Warehouse

“Through measurement comes knowledge.”

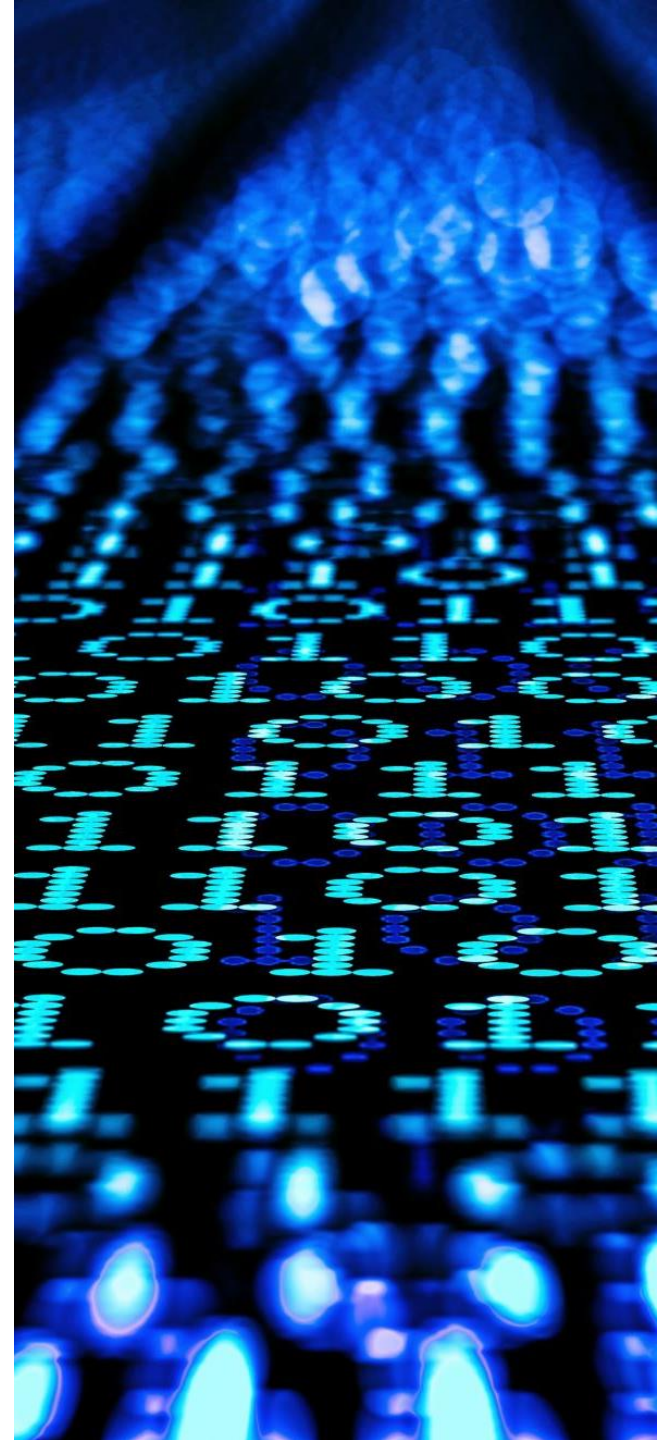
Heike Kamerlingh Onnes

A data warehouse is an example of the journey that data takes, when combined with context, to become information.

Prior to application of context it is just a collection of numbers and letters, bits and bytes.

Yet information is still not enough to enable an organization to learn from and act based on what they have collected.

The ability to use accurate data and timely information to objectively measure and, therefore, proactively manage outcomes and business processes demonstrates the value of a data warehouse.



Data Warehouse?

How do you understand the notion –
Data Warehouse?

What is it?

*System / application / architecture /
technique / database*

Purpose – why we use it?

essence and basic characteristics ?

Components -

any ? Types ?

Data

types and characteristics ?

Processes

types and characteristics ?

Data Warehouse – as a data repository

A dedicated database system for decision making

***separate** from the production database(s) used operationally*

Differs from production system in that:

*it covers a much **longer time horizon** than transaction systems*

*it includes multiple data bases that have been **processed** so that the warehouse's data are defined uniformly (i.e., 'clean' data)*

*it is **optimized** for answering complex queries from managers and analysts.*

Data warehouse – definitions

Definitions:

“A **read-only analytical database** that is used as the foundation of a decision support process”

(Poe and Reeves, 1995)

“An **integrated and consistent store of subject-oriented data** that is obtained from a variety of sources and formatted into a meaningful context to support decision-making in organization”

(McFadden, 1999)

“Managed data situated **after and outside the operational system**”

(Gupta 1997)

„A **copy of transaction data** specifically structured for query and analysis”

(Kimball, 1996)

What is not a data warehouse

If we read again the above definitions, we can see they both **define a data warehouse as a kind of a data collection**

It is not a process of collecting, transforming or analysing the data, not a tool for data integration, neither it is a reporting tool.

It is a data storage, and this is the key property.

But **not every data storage is a data warehouse.**

This data must serve a specific purpose.

Even a transactional database, holding all the transactional information in a neatly designed 3NF, does not fulfill these conditions.

DW prefix is not enough

Not just a view with multiple unions

Not a dumping ground

Data warehouse – definitions

Data warehouse is an architecture, the entire process focused on these particular tasks, which requires multiple tools and separate approaches

“A copy of transaction data specifically structured for query and analysis. (..) A data warehouse is a system that extracts data from source systems, transforms and loads to multidimensional structures, and further supports queries and reporting for decision support.”,

Kimball

“A data warehouse is a storage architecture designed to hold data extracted from transaction systems, operational data stores, and external sources. The warehouse then combines that data in an aggregated, summary form suitable for enterprise-wide data analysis and reporting for predefined business needs.”

Gartner Glossary, <https://www.gartner.com/en/information-technology/glossary/data-warehouse>

Data Warehouse

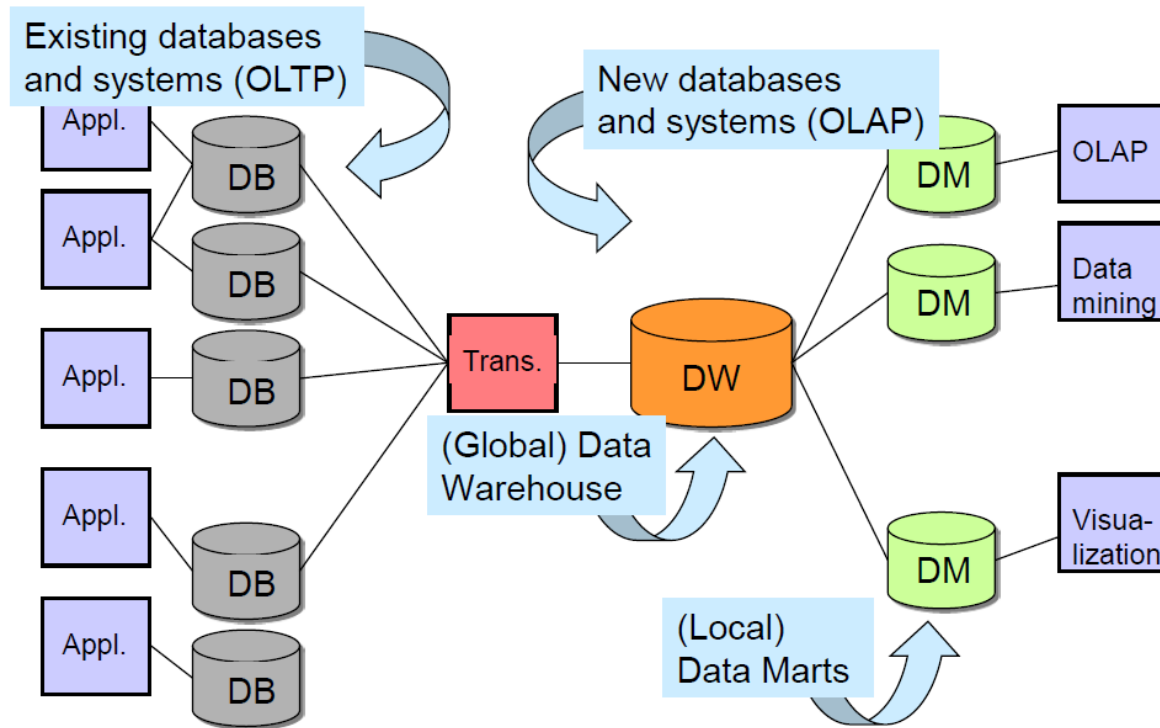
Data warehousing:

The entire process involved in constructing and using data warehouses

It consists of ?

It requires ?

Data Warehouse – as an architecture



Analogy: (data) producers ↔ warehouse ↔ (data) consumers

The Definition

The most common definition of Data Warehouse

A data warehouse is a **subject-oriented, integrated, time-variant**, and **non-volatile** collection of data in support of management's decision-making process

W.H. Inmon, 1992

Data Warehouse

Subject oriented:

In operational systems

Data is stored and organised around individual applications; data sets contain data needed only for this particular application

Operational database focuses on support of a particular business process

E.g. Handling appointments and bookings

Function oriented

In data warehouse

Data is stored by real-world business subjects or events, not by application; focuses on analysis of particular subjects; data is linked and stored by business subjects

E.g. Products

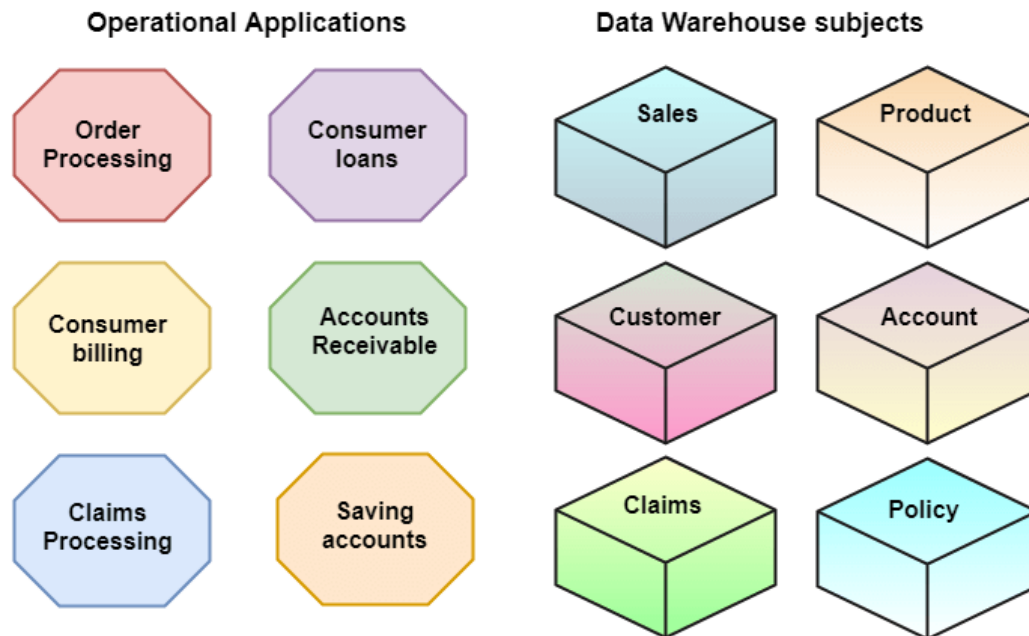
In a data warehouse, there is no application flavor

data in a data warehouse cuts across applications

Subject oriented

Data Warehouse

Data warehouses typically provide a concise and straightforward view around a particular subject (customer, product, sales, etc.), instead of the global organization's ongoing operations (order processing, warehouse management, etc.)



Data Warehouse

Subject oriented:

- oriented towards the major subject areas of the corporation
defined in the data model and information needs

- organized around major subjects

 - focusing on the modelling and analysis of data for decision makers*

- focused on specific business analyzes

 - Data is collected for and organized in such a way as to facilitate these specific analytical tasks*

- provide a simple and concise view

 - around particular subject issues*

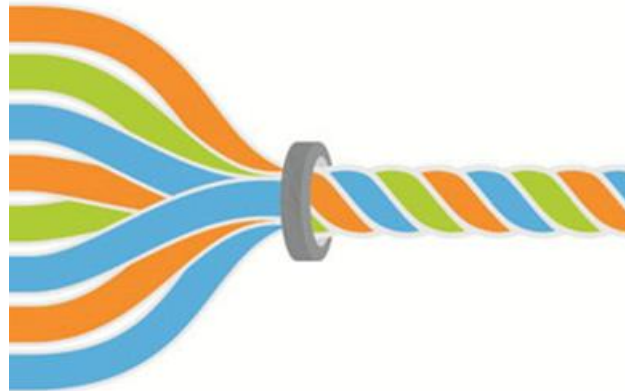
 - excluding data that are not useful in the decision process*

Data Warehouse

Integrated:

create a conformed and uniform view of the subject data

pulling together all the relevant data from various applications; inside and outside to the organization



Data Warehouse

Integrated:

constructed by integrating multiple, heterogeneous data sources

relational databases, flat files, ...

data cleaning and data integration techniques are applied

there is not consistency among different data sources

heterogeneous data sources

when data is moved to the warehouse, it is processed

ensure consistency

when data is moved to the warehouse, it is processed

Data Warehouse

Integrated:

Data cleaning and data integration techniques are applied

in naming conventions

e.g., LastName and FamilyName in DB1 and DB2 have the same signification

in encoding structures

e.g., Attribute User_Id is a long int in DB1 and it is a string in DB2

in attribute scales

e.g., cm vs inch

etc.

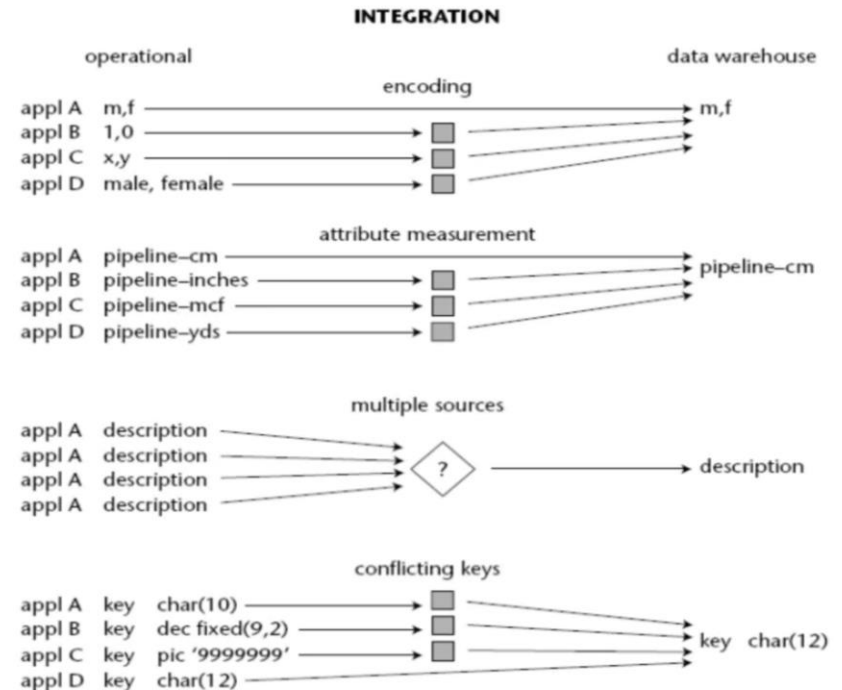


Figure 2-2 The issue of integration.

Data Warehouse

Integrated:

Logic organisation

Semantic organisation

Dictionaries are aligned

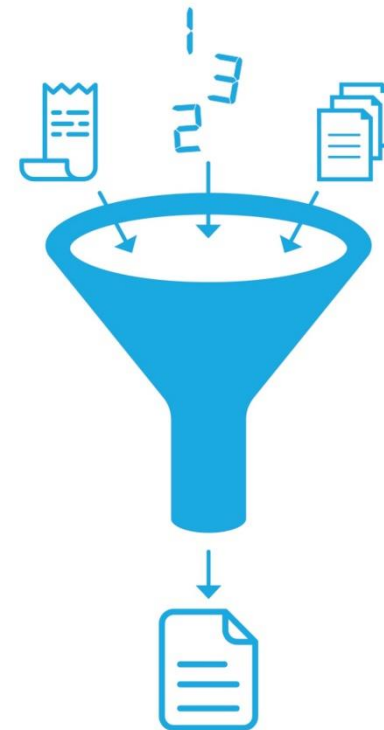
Provide a single, detailed, and consistent view

Resolve conflicting data

Single-version of truth

Physical organisation

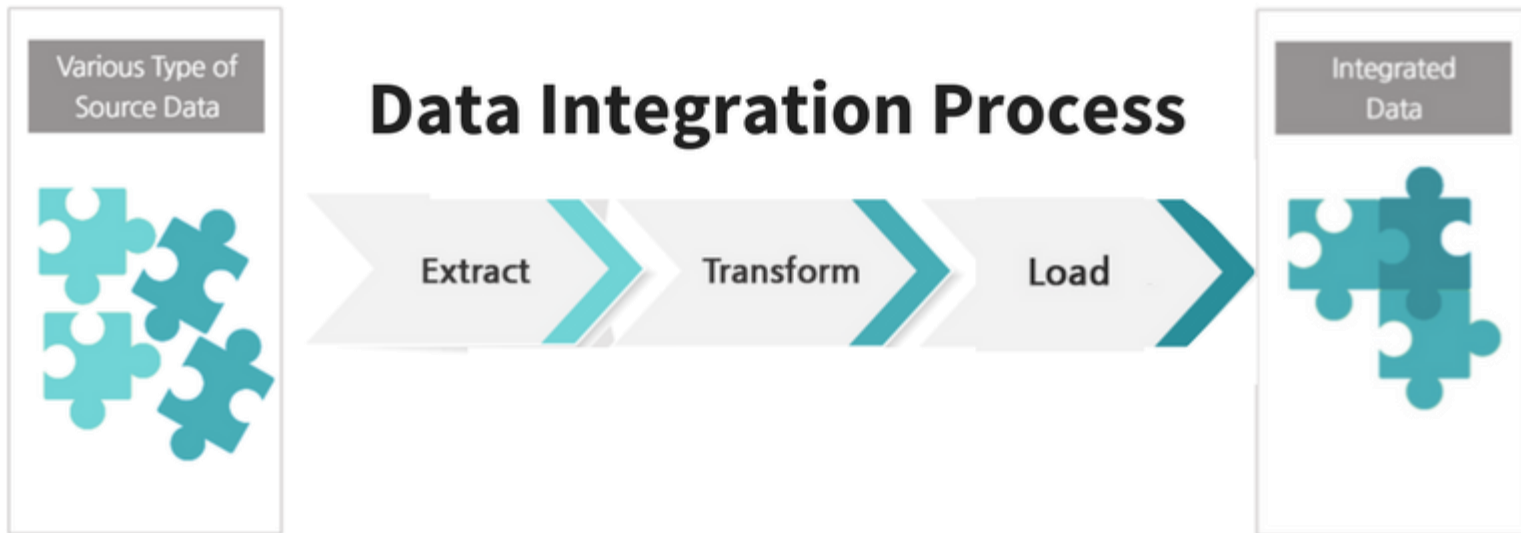
Format, data types, etc.



Data Warehouse

Having a DW

It is the first time the company has an integrated view of its information



Data Warehouse

Non-volatile:

- a physically separate store

- warehouse data is loaded and accessed

- typical database update of data does not occur in the data warehouse environment

 - does not require transaction processing, recovery, and concurrency control mechanisms*

- requires only two operations in data accessing:

 - loading of data (governed write) and querying (read)*

- data is frozen

Data Warehouse

Non-volatile:

- a physically separate store
- warehouse data is loaded and accessed;

- update of data does not occur in the data warehouse environment*

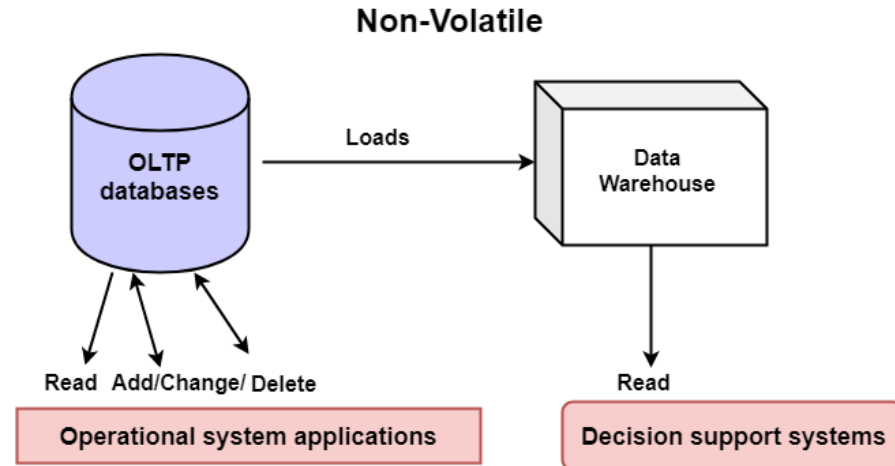
- operational update of data does not occur in the data warehouse environment.

- Does not require transaction processing, recovery, and concurrency control mechanisms*

- Requires only two operations in data accessing:*

- initial loading of data and querying (read)

- Data is frozen



Data Warehouse

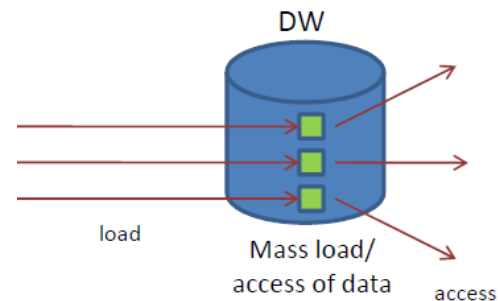
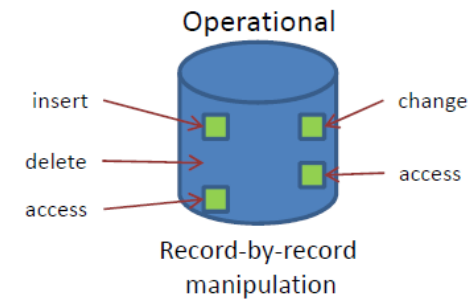
Non-volatile:

Once placed in the warehouse, data are usually not subject to changes

other than adding new portions of data

Every database user can be sure that the query will always return the same result

no matter how often it is executed



Data Warehouse

Time-variant:

In operational system

the stored data contains the current values.

In an accounts receivable system, the balance is the current outstanding balance in the customer's account.

we typically store some past transactions in operational systems

essentially, operational systems reflect current information because these systems support day-to-day current operations.

In data warehouse

data is accurate and correct only in reference to a particular point of time

Analysing buying pattern of a specific customer we need data, not only, about the current purchase, but on the past purchases as well

because of the very nature of its purpose, has to contain historical data

data is stored as a series of snapshots over past and current periods

each snapshot can represent a point/period of time

changes are tracked and recorded

every data element in the data warehouse contains an element of time – explicitly or implicitly

as opposed to operational database that typically maintains no historical data

maybe just a specific type, for different purposes

this aspect of the data warehouse is quite significant for both the design and the implementation phases.

Data Warehouse

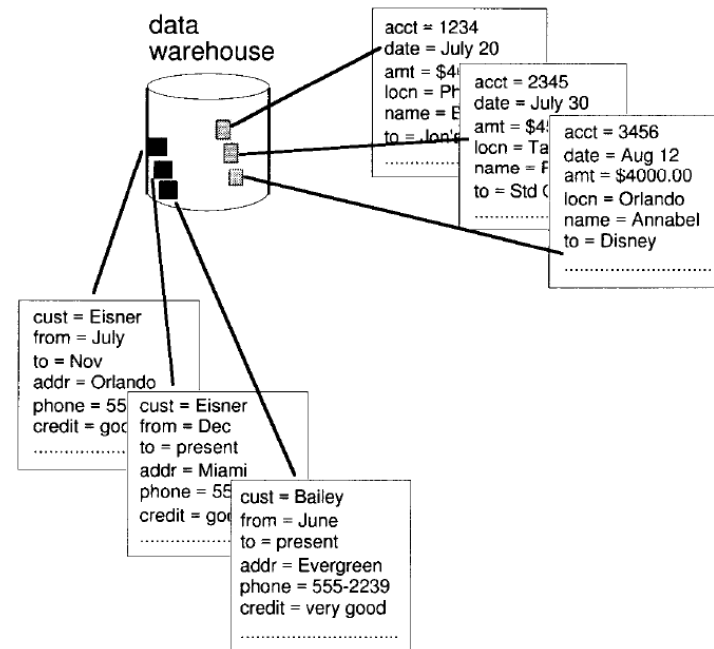
SNAPSHOTS

Data in the data warehouse is stored in units of "snapshots".

The records in the data warehouse are

created as of some moment in time

created in effect a snapshot taken as of that moment in time



Data Warehouse

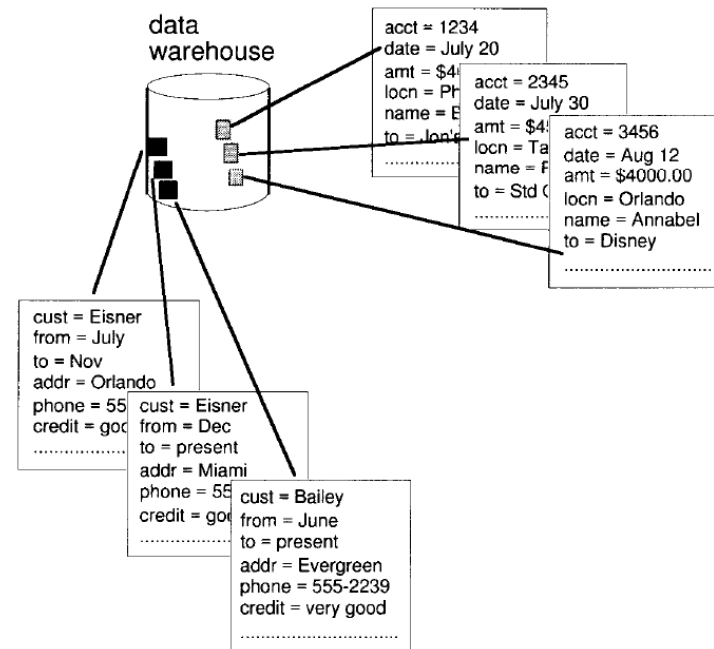
SNAPSHOTS

In this regard the data in the data warehouse is fundamentally different from the data in an operational data base environment

Data in an operational data base environment can be updated.

Since data in the data warehouse environment is snapshot data it cannot be updated

Are „frozen”



Data Warehouse

There are many different forms of taking snapshots

The most basic consideration of a snapshot is that the snapshot has been taken as a result of an event.

The event may be triggered by a wide variety of occurrences:

*an occurrence of a transaction,
the periodic passage of time,
a threshold having been reached,
an audit,
a special request, etc.*

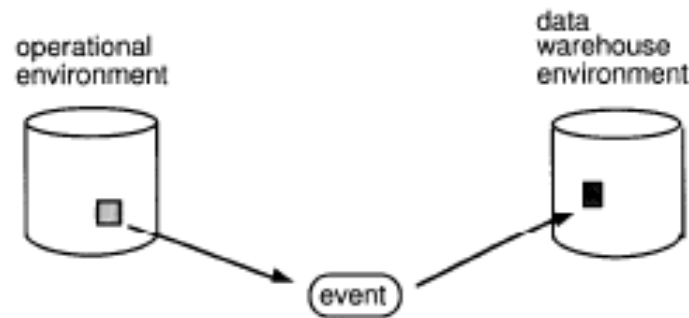


Figure 2: Data is moved into the data warehouse as a result of the occurrence of some event.

Data Warehouse

What exactly is a process?

A business process refers to a wide range of structured, often chained, activities or tasks conducted by people or equipment to produce a specific service or product for a particular user or consumer

The entity-relationship model is used to describe information

the entity-relationship guides the database design of an operational system,

The process model is used to describe business activity

the process model guides design of the functional components.

In short:

process models involve functional decomposition

DW works

by capturing details about significant events or transactions, it constructs a record of activities

DW focus is on:

modelling the measurement of business processes

support the analysis of business processes

not supporting the execution of business processes – OLTP

Data Warehouse

The snapshot triggered has four basic components:

A key

identifies the record and primary data

A unit of time

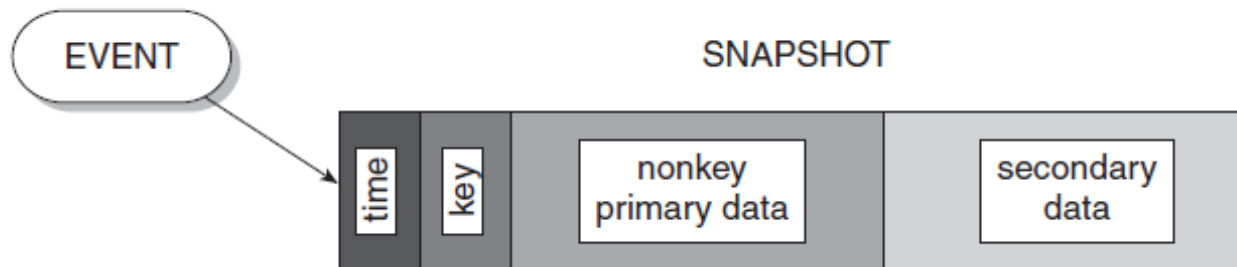
usually refers to the moment when the event occurred

Primary data that relates directly to the key

Secondary data captured as part of the snapshot process that has no direct relationship to the primary data or key

incidental data that might be later used to support decisions

extraneous information captured at the moment of snapshot



Data Warehouse

Example:

A key

identifies a sales of product

A unit of time

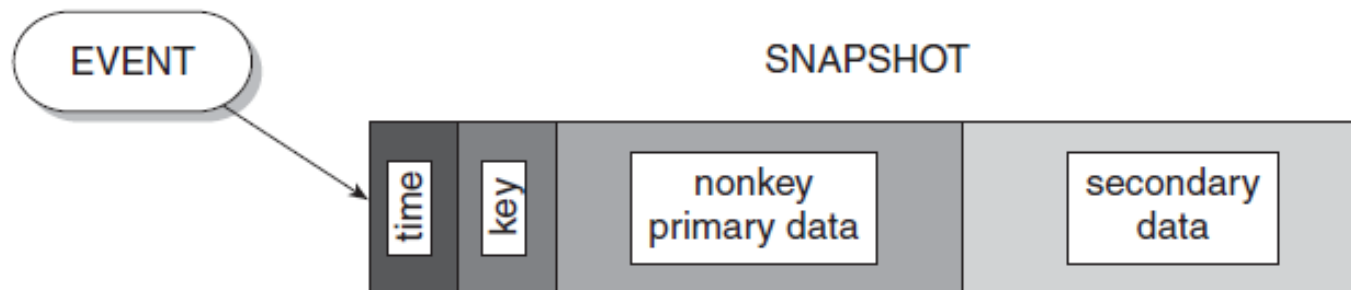
identifies when this sales happened

Primary data that relates directly to the key

identifies product, price, location, etc.

Secondary data

how much product is in stock, interest rate



Data Warehouse

Time-variant:

- the time horizon for the data warehouse is significantly longer than that of operational systems

 - provide information from a historical perspective*

 - e.g., past 5-10 years

- data in the data warehouse are not static

- data is stored as a series of snapshots

 - each snapshot represents a period of time*

- every data in the data warehouse

 - contains an element of time, explicitly or implicitly*

 - but the data of operational database may or may not contain “time element”*

Data Warehouse

Time-variant:

- incremental data loads

 - changes invoke data addition and not data updates*

- the time horizon for the data warehouse is significantly longer than that of operational systems

 - provide information from a historical perspective*

 - e.g., past 5-10 years

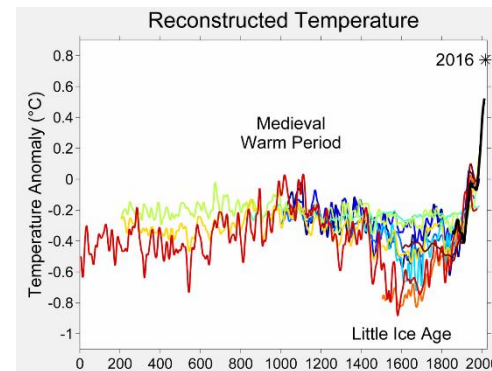
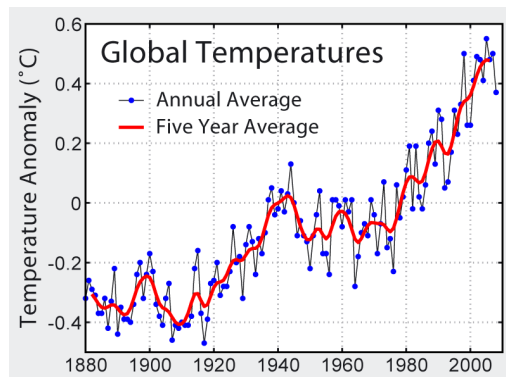
Data Warehouse

Having a DW

First opportunity to store history – a robust amount of history

History is an invaluable in understanding your company's

Customers, trends, products, etc.



Data Warehouse

Consider that a particular health clinic changed location (address)

How is this handled in operational database ?

How should this be handled in a data warehouse environment ?

Non-volatile and time-variant?

Is it possible?

Data Warehouse

Non-volatile and time-variant ?

Consider that a particular health clinic changed location (address)

How this is handled in operational database ?

How should this be handled in a data warehouse environment ?

Data warehouse

Data granularity

In an operational system

data is usually kept at the lowest level of detail.

In a POS system, the units of sale are captured and stored at the level of units of a product per transaction at the check-out counter.

In an order entry system, the quantity ordered is captured and stored at the level of units of a product per order received from the customer.

whenever you need summary data

you add up the individual transactions.

If you are looking for units of a product ordered this month, you read all the orders entered for the entire month for that product and add up.

you do not usually keep summary data in an operational system.

Data warehouse

Subject-oriented:

Data in an organization is organized in major objects or business processes.

The data should be organised around subjects they describe or refer to, rather than by systems they originate from.

Integrated:

Data warehouse integrates data from various sources across departments within the organization. Different data sources can have different ways to define a specific objects – in a data warehouse, there must be only one definition.

Non-volatile:

Data in the data warehouse is not subject to change. Once placed in a data warehouse, data is not updated (to some point).

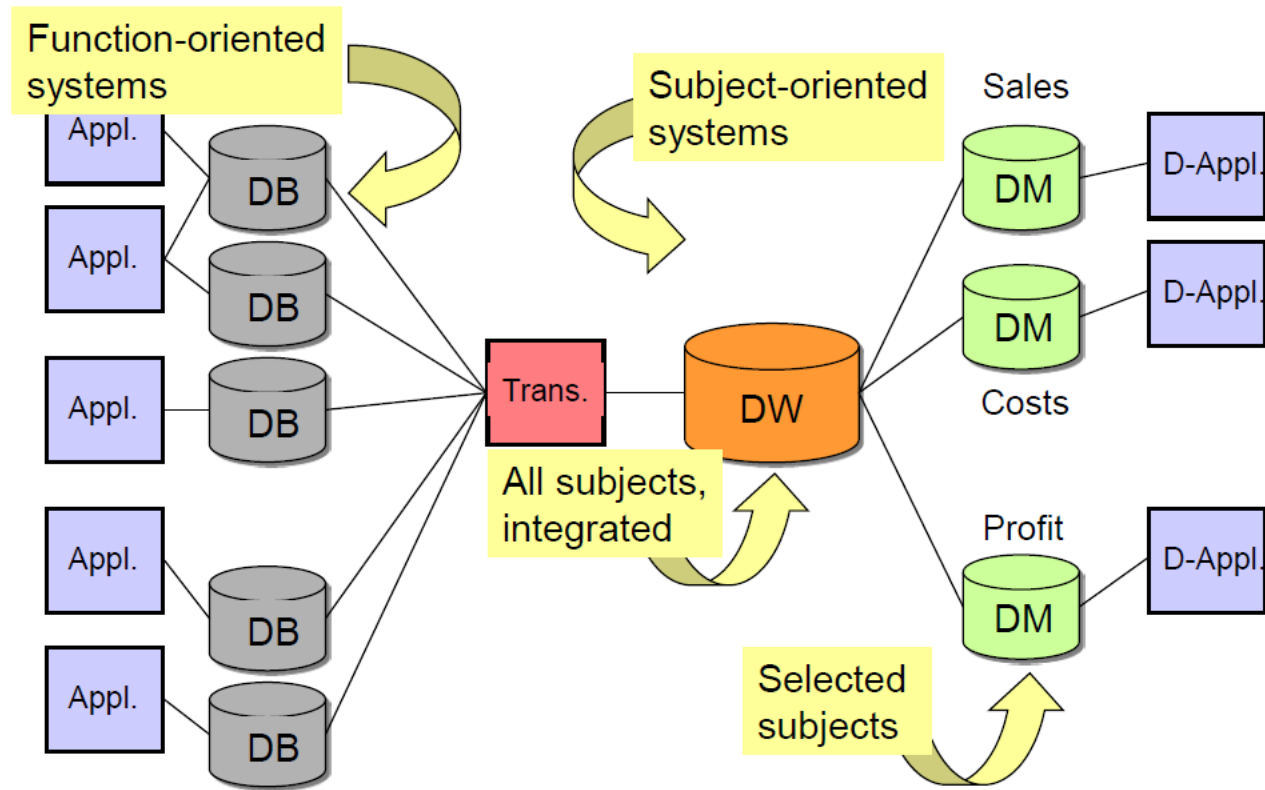
A permanent storage of the data

Time-variant:

Data in data warehouse associates with time. The time can be a single moment or span of time.

For the data warehouse to serve its purpose its content must be regularly and frequently updated.

Data warehouse



Data warehouse

Data granularity

When a user queries the data warehouse for analysis, he or she usually starts by looking at summary data.

The user may start with total sale units of a product in an entire region.

Then the user may want to look at the breakdown by states in the region.

The next step may be the examination of sale units by the next level of individual stores.

Frequently, the analysis begins at a high level and moves down to lower levels of detail.

In a data warehouse

you find it efficient to keep data summarized at different levels.

Depending on the query, you can then go to the particular level of detail and satisfy the query.

data granularity in a data warehouse refers to the level of detail.

The lower the level of detail, the finer is the data granularity

Data warehouse

A data mart (DM)

many people confuse these terms (data warehouse)

Even some authors and vendors use the two terms synonymously

is a simple form of a data warehouse that is focused on a single subject (or functional area),

such as Location, Sales, Finance, or Marketing

single LOB – line of business

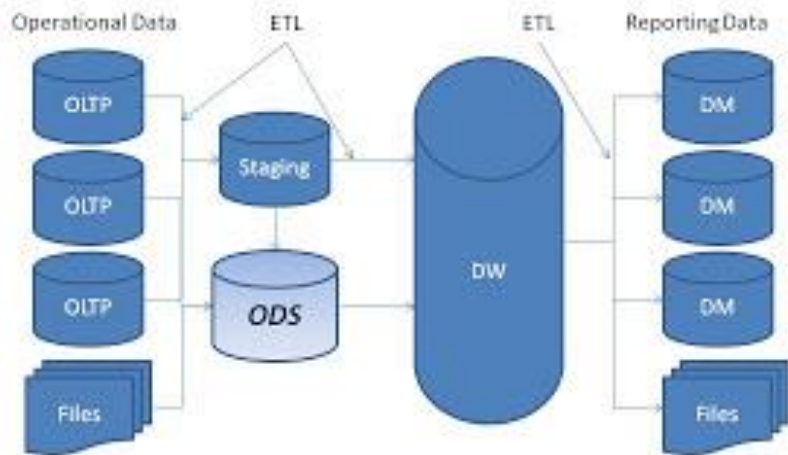
given their single-subject focus, data marts usually draw data from only a few sources

the sources could be internal operational systems, a central data warehouse, or external data

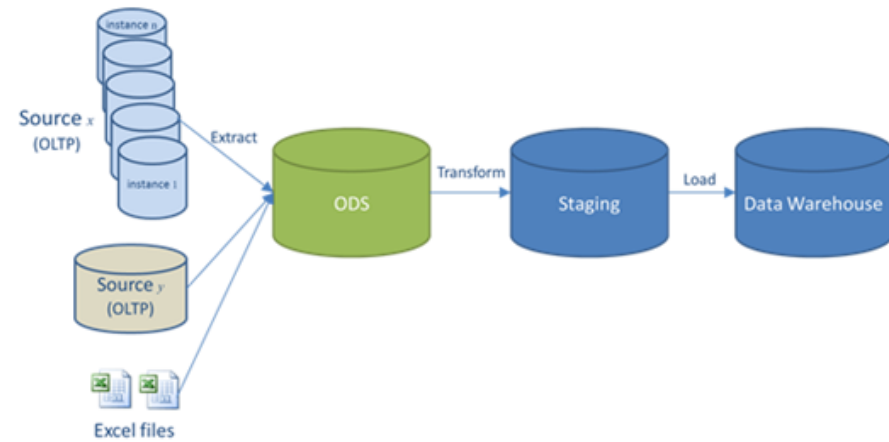
A data warehouse, unlike a data mart, deals with more complex and general data

Data warehouse

ODS within Data Warehouse Architecture



<http://randygrenier.blogspot.com/>



<https://www.iri.com/>

Data warehouse

An operational data store (ODS)

a central database that provides a snapshot of the latest data from multiple transactional systems for operational reporting

enables organizations to combine data in its original format from various sources into a single destination to make it available for business reporting.

ODS is not a data warehouse or data mart.

The purpose of an ODS is to integrate corporate data from different heterogeneous data sources in order to facilitate operational reporting in real-time or near real-time.

Usually data in the ODS will be structured similar to the source systems, although during integration the data can be cleaned, denormalized, and business rules applied to ensure data integrity.

This integration will happen at the lowest granular level and occur quite frequently throughout the day.

An ODS is targeted for the lowest granular queries

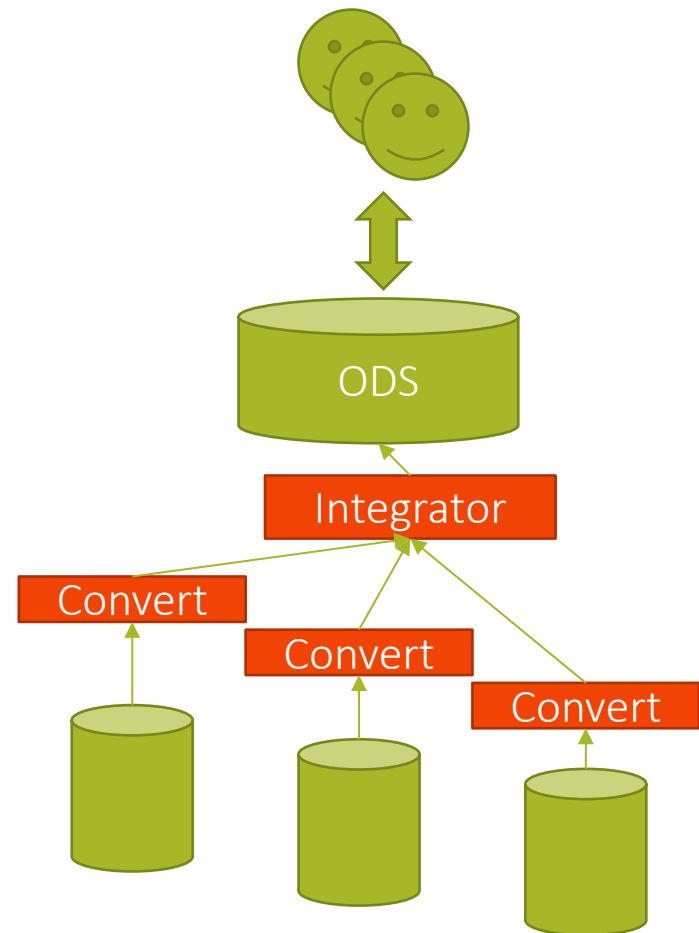
Normally an ODS will not be optimized for historical and trend analysis

Operational Data Store (ODS)

Because of limitations suffered by on demand and eager approaches, it led to the development of the Operational Data Store (ODS)

Operation Data Store is a special unified (separate) database that has been developed specifically to handle management queries.

a short-term enterprise memory which contains recently stored organizational information.



Operational Data Store (ODS)

Inmon and Imhoff (1996) defined ODS or Operational Data Store as

‘A subject-oriented, integrated, volatile, current valued data store, containing only corporate detailed data.’

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‘A subject-oriented, integrated, volatile, current valued data store, containing only corporate detailed data.’

Subject-Oriented

Built around the major data entity or subjects of an organization.

For example, in a University, the data subject (entities under study) could be students, teachers, and various courses, whereas in a company the subjects might be workers, interns, services and items.

Operational Data Store (ODS)

Inmon and Imhoff (1996) defined ODS or Operational Data Store as

‘A subject-oriented, integrated, volatile, current valued data store, containing only corporate detailed data.’

Integrated

integrates (combines) data from multiple systems to provide a single point of wide view of any enterprise's data.

Operational Data Store (ODS)

Inmon and Imhoff (1996) defined ODS or Operational Data Store as

‘A subject-oriented, integrated, volatile, current valued data store, containing only corporate detailed data.’

Current valued

Focused on up to date as it doesn't contain obsolete data – always current valued due to its information nature

information is always up to date and shows the current state

Operational Data Store (ODS)

Inmon and Imhoff (1996) defined ODS or Operational Data Store as

‘A subject-oriented, integrated, volatile, current valued data store, containing only corporate detailed data.’

Volatile

a short term enterprise memory due to its volatile nature – updates its information, hence its data is volatile in nature

the older information is always replaced by the updated information

Operational Data Store (ODS)

Inmon and Imhoff (1996) defined ODS or Operational Data Store as

‘A subject-oriented, integrated, volatile, current valued data store, containing only corporate detailed data.’

Detailed

detailed enough to cater to the needs of the operational management staff in an enterprise.

Although it does not have to be as descriptive as its source OLTP is (subject-oriented), it is still capable enough to highlight important organizational information.

Operational Data Store (ODS)

Types of ODS

An ODS could be classified into different types depending upon the needs of an organization

Type I: ODS as an operational reporting tool

This type of ODS is regularly updated, typically daily, making it the best tool for operational reporting on organization's day-to-day operations and activities.

These reports typically include data pertaining to production costs, accounts, resource expenditures, and comprehensive examinations of processes.

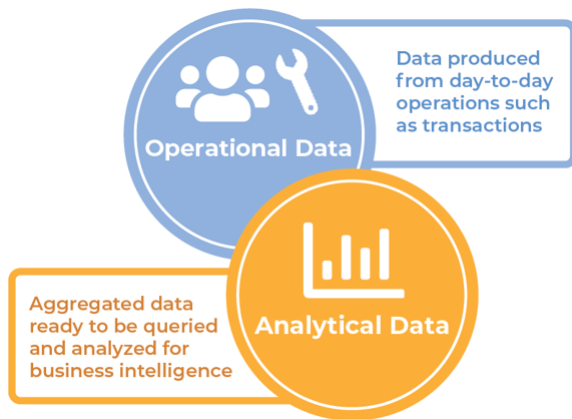
For example, it could be used for providing total sales made per day, number of transactions done in a day and other logs.

Type II: ODS as a complex information tracker

Complex information such as locations, products, production duration can be easily derived using ODS. This kind of database is updated more frequently (perhaps hourly) to reflect changes.

Type III: ODS as customer relation management supporter

fed with synchronous or near-synchronous updates to provide customers with the most accurate and up to date information.



Operational reporting

gives a tactical and a structural view of an organization and provides details about the daily aspects of operations and business intelligence insights that are instantly actionable

For example, current inventory levels or sales values

use this information to make operational decisions and handle day-to-day operations

Analytical reporting

sets and meets (or exceeds) organizational goals on a long-term basis by using qualitative and quantitative data to analyze and evaluate a business strategy or process

For example, trends in inventory levels or sales values

use the information to make strategic decisions and pivot on their strategy

The ODS can be extended further to build a data warehouse which can store historical information as well:

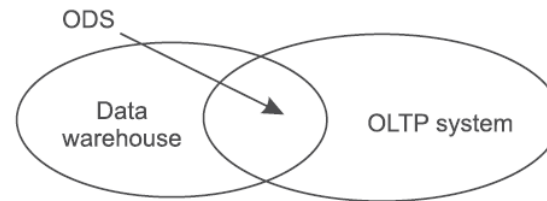


Figure 12.2 Relationship between OLTP, ODS and data warehouse systems

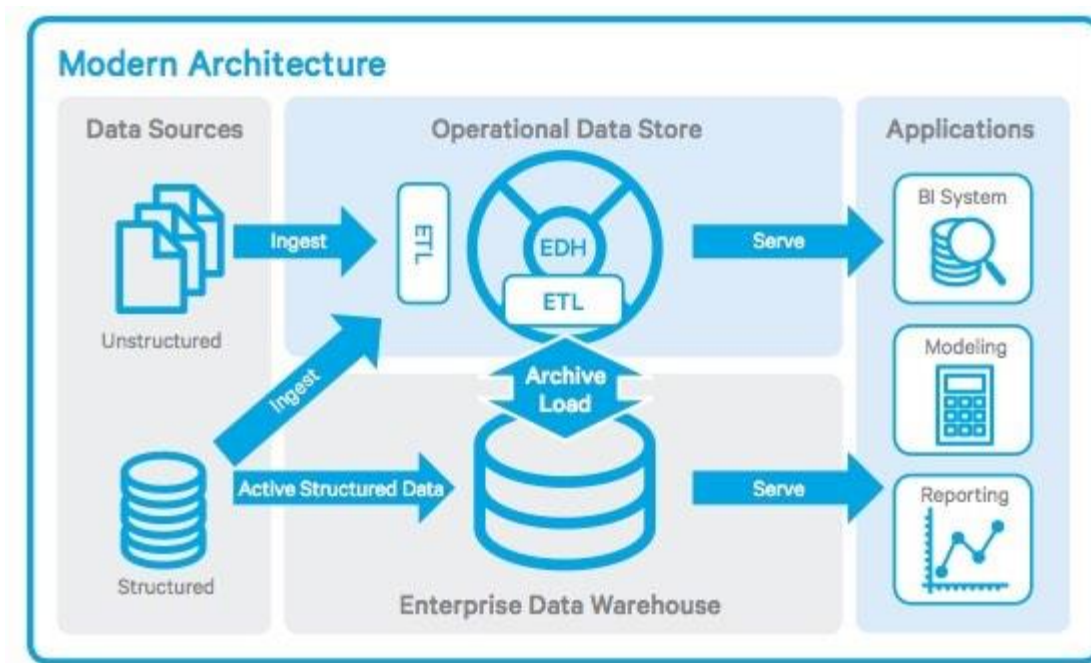
Data warehouse

Operational Data (TP)	Operational Data Store (ODS)	Data Warehouse	Data Mart
Ranges from application-oriented to subject oriented	Subject-oriented within solution domain	Subject-oriented	Subject-oriented within application domain
Ranges from non-integrated to integrated	Integrated within solution domain.	Integrated	Integrated within application domain
Current, may have limited history and/or archiving	Current or near current	Time-variant	Time-variant
Volatile, continuously updated	Volatile and updateable	Non-volatile	Non-volatile

Data warehouse

Operational Data (TP)	Operational Data Store (ODS)	Data Warehouse	Data Mart
Ranges from application owned to enterprise managed	Solution focused	focused	Application focused
Optimized for transaction processing	Optimized for OLTP or data access	Optimized for distribution, query, and reporting	Optimized for decision support, analysis
Ranges from application specified to architected	Designed as a local or interim solution	Architected as long-term information asset	Architected as a point solution to a business need
Most commonly driven by a legacy requirements.	Driven by scope of legacy requirements.	Implemented with an evolutionary approach	Rapidly deployed and re-deployed

Data warehouse

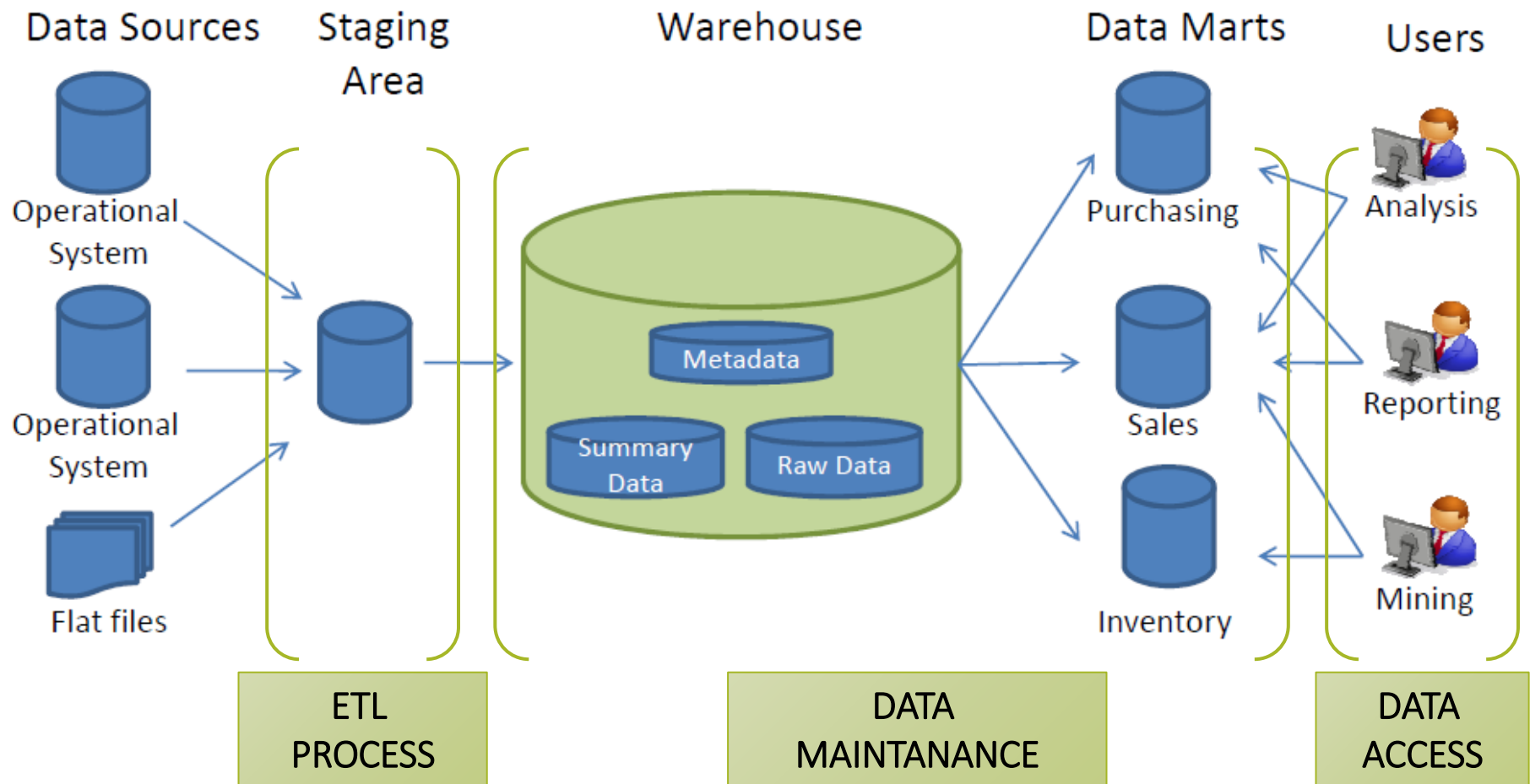




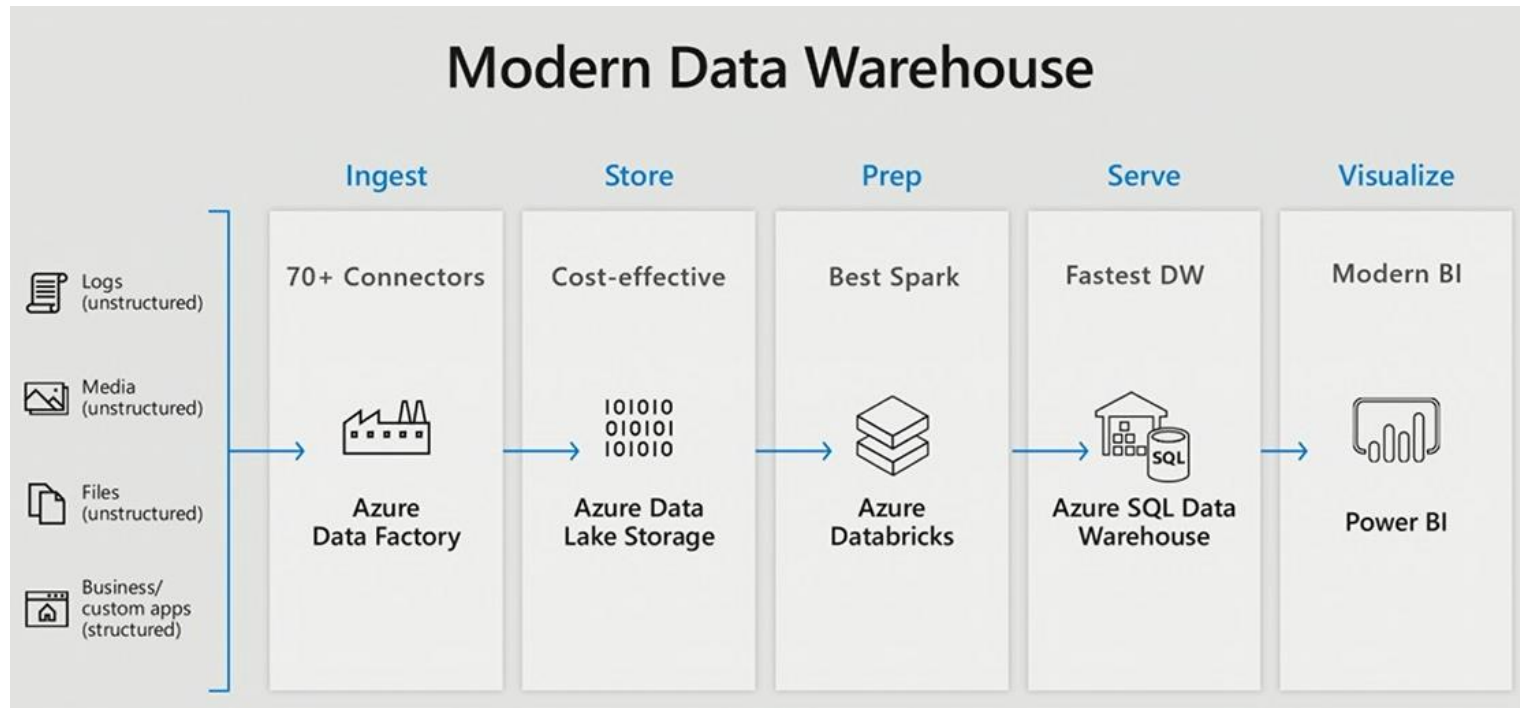
Data Warehousing

The process

Processes



Process

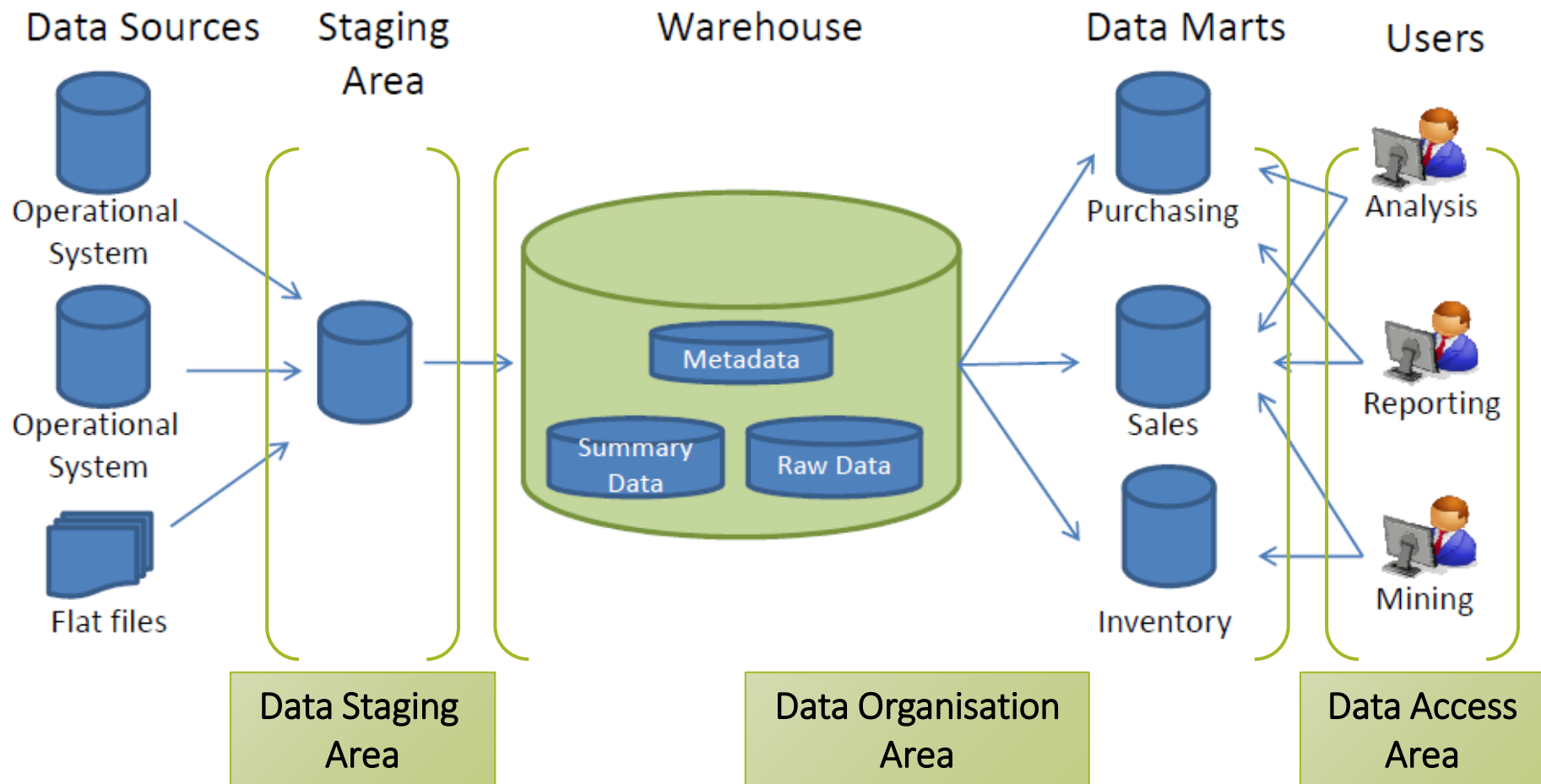


A close-up photograph of server hardware, showing two glowing green indicator lights on a dark blue or black metal frame. The lights are slightly out of focus, creating a bokeh effect. The background is a solid, vibrant green color.

Data Warehousing

The data

Logical Components



Data Staging area

Data Staging Area

is both a storage and process area (the ETL process)

it represents everything that happens between the operational source system and the data organisation area

input

Unintegrated, operational environment ('legacy systems')

the key architectural requirement for data staging area is that

it is off-limits to business users

does not provide query and presentation services

it is acceptable to create a normalized database to support the staging processes

it is not the end goal

Normalized structures must be off-limits to user queries

because they defeat understandability and performance

Data Staging area

Handle ETL

Data extraction

get data from multiple, heterogeneous, and external sources

Data cleaning

detect errors in the data and rectify them when possible

Data transformation

convert data from legacy or host format to warehouse format

Load

sort, summarize, consolidate, compute views, check integrity, and build indices and partitions

Refresh

propagate the updates from the data sources to the warehouse

Data Organisation Area

Data Organisation Area:

is the data warehouse as far as the business community is concerned

data is organized, stored, and made available for direct querying by users, report writers, and other analytical applications

holds both aggregated and detailed data for management

separate from the databases used for OLTP

data is based on a multidimensional data model

views data in the form of a data cube

Data Organisation Area

Goals

- Ensure an integrated view over data

 - Entire set of data available in the enterprise*

- Provide technologies for analytical processing – OLAP

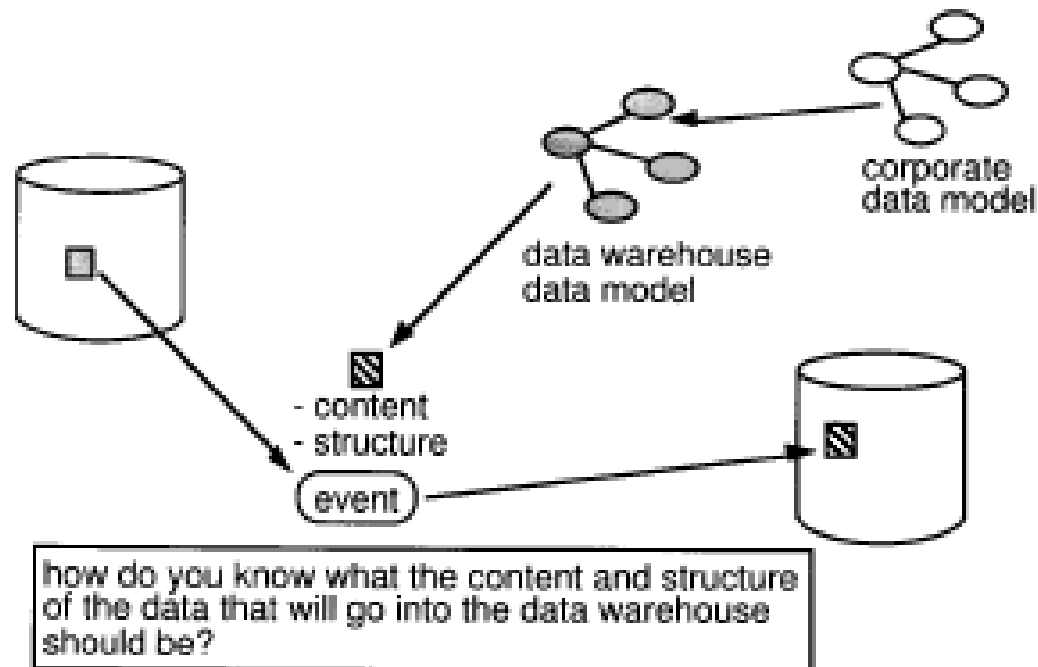
 - Allow for advanced analytical querying*

 - Allow for data mining*

 - Discover hidden patterns and trends

 - Ad-hoc analysis and reports*

Data in Data Warehouse



Data Access

Data Access Area:

variety of capabilities can be provided to business users to leverage the presentation area for analytic decision making

pivot tables/charts

tabular view

data mining structure

all data access tools in the data warehouse's presentation area query the data DBMS (organisation area)

data access tools can be

as simple as an ad hoc query tool

as complex as a sophisticated data mining or modelling application

Data Access

On-Line Analytical Processing (OLAP)

- Interactive analysis

- Explorative discovery

- Fast response times required

OLAP operations/queries

- Aggregation, e.g., SUM

- Change level, e.g. (Year, City) -> (Year, Month, City)

 - Roll Up: Less detail*

 - Drill Down: More detail*

- Filtering, e.g. Year=2000

 - Slice/Dice: Selection,*

DW Environment

Performance Optimization

The data warehouse contains GBytes or even TBytes of data!

OLAP users require fast query response time

They don't want to wait for the result for 1 hour!

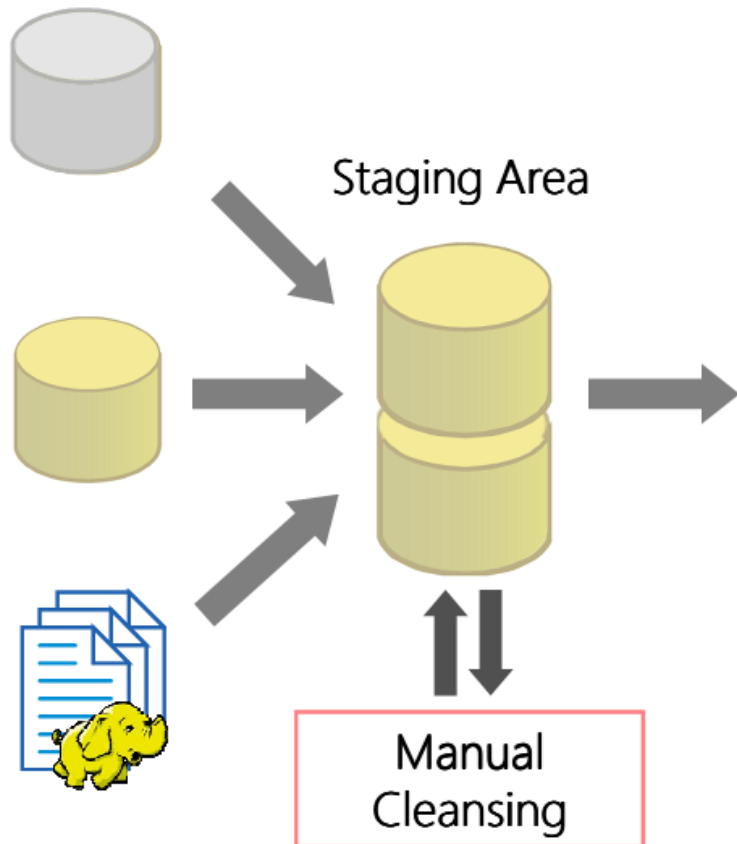
Acceptable: answer within seconds, maximum minutes

Idea:

?

DW Environment

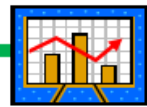
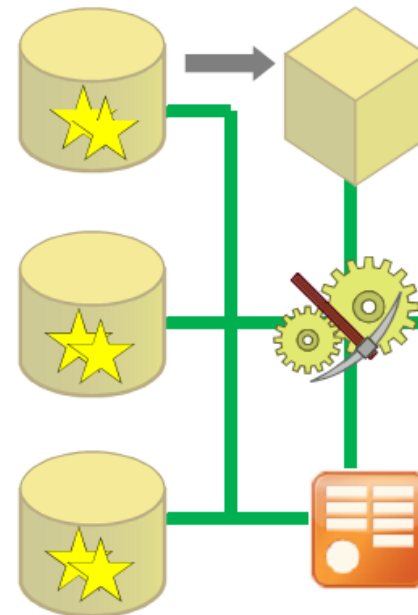
Data Sources



Data Warehouse

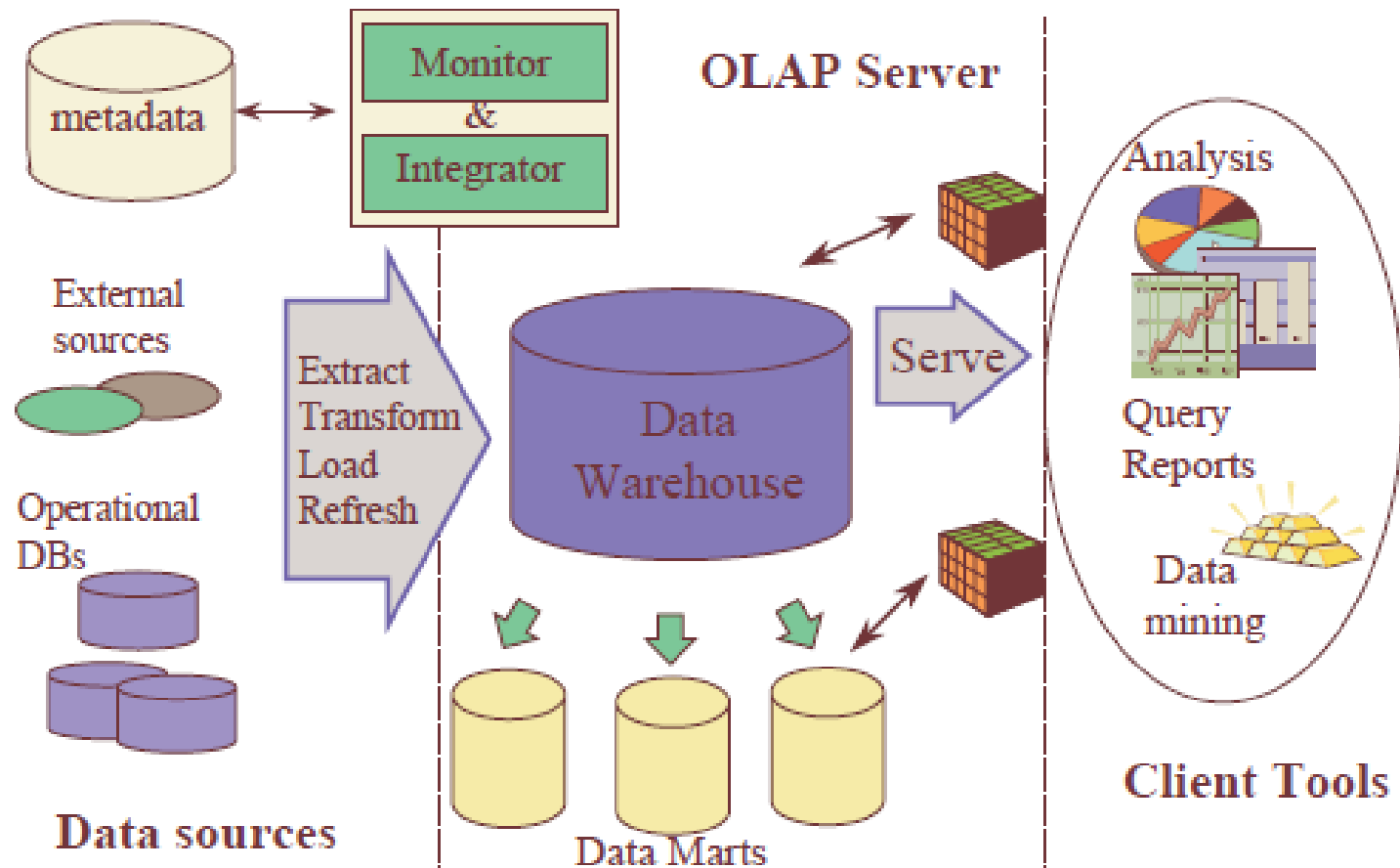
Business Rules

Data Marts



Client
Access

DW Environment



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Bibliography

[SQL Tutorial: OLTP and OLAP – YouTube](#)

OLTP / OLAP

https://www.youtube.com/watch?v=AHR_7jFCMeY

DW very brief, general information

<https://www.youtube.com/watch?v=k4tK2ttdSDg>

DW general information, with some details on deployment approaches

<https://www.youtube.com/watch?v=TZHykX6cEyc>

Modern data warehouses, within MS ecosystem

<https://www.youtube.com/watch?v=eOQ3YJKgvHE>

Modern data warehouse, within Google ecosystem

https://www.youtube.com/watch?v=Tff34jj_V-0

Inmon / Kimbal architectures

Some nice
short videos